

## Unit 9

# ELECTRICITY

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55 items: 22 from official papers (115 marks), 33 from the booklet

### **Syllabus sections in this unit**

9.1 Electric current

9.2 Potential difference and power

9.3 Resistance and resistivity

## Unit 9: Electricity

### 1. M/J 26/P22/Q5/b(ii)

- 5 (a) Three resistors of resistances  $R_1$ ,  $R_2$  and  $R_3$  are connected in a network in parallel as shown in Figure 5.1.

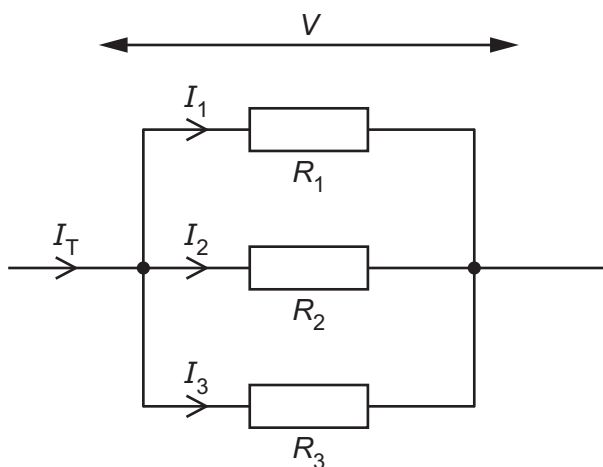


Figure 5.1

The currents in the resistors are  $I_1$ ,  $I_2$  and  $I_3$  respectively. The total current entering the network is  $I_T$ . The potential difference (p.d.) across the resistors is  $V$ .

Derive the equation

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$$

where  $R_T$  is the total resistance of the network.

[2]

- (b) Two identical resistance wires, each of resistance  $2.8\Omega$ , are connected in parallel to a battery with negligible internal resistance and electromotive force (e.m.f.)  $12\text{V}$  as shown in Figure 5.2.

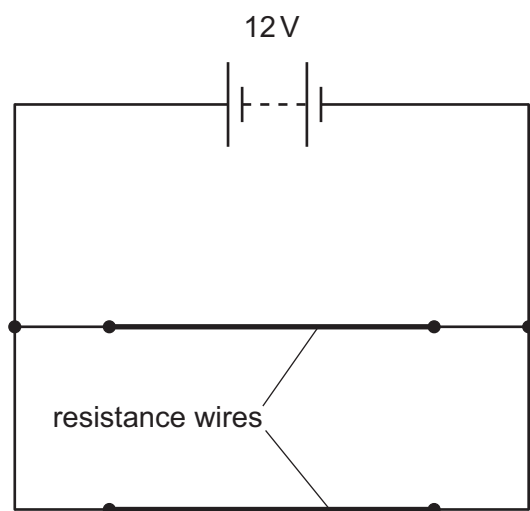


Figure 5.2

- (ii) Each resistance wire has a cross-sectional area of  $6.1 \times 10^{-6}\text{m}^2$ .

The wires are made from a metal with a number density of charge carriers (free electrons) of  $8.5 \times 10^{28}\text{m}^{-3}$ .

Calculate the average drift speed of the charge carriers in the wires.

$$\text{speed} = \quad \text{ms}^{-1} \quad [2]$$

**2. M/J 26/P23/Q4/b(ii,iv)**

- 4** A light-dependent resistor (LDR) is connected across the terminals of a cell that has internal resistance. A voltmeter is connected across the LDR.

**(b)** The cell has an electromotive force (e.m.f.) of 4.90 V.

At a particular light intensity, the LDR has a resistance of  $230\,\Omega$  and a potential difference (p.d.) across it of 4.78 V.

**(ii)** Calculate the power dissipated in the LDR.

$$\text{power} = \quad \text{W} \quad [2]$$

**(iv)** The intensity of the light illuminating the LDR is gradually decreased.

By placing **one** tick (✓) in each row, complete Table 4.1 to show what happens to the quantities indicated as the light intensity decreases.

**Table 4.1**

|                          | decreases | unchanged | increases |
|--------------------------|-----------|-----------|-----------|
| <b>resistance of LDR</b> |           |           |           |
| <b>current in cell</b>   |           |           |           |
| <b>p.d. across LDR</b>   |           |           |           |

[3]

**3. M/J 26/P24/Q5/a,b,d**

- 5** The  $I$ – $V$  characteristic of an electrical component X is shown in Figure 5.1.

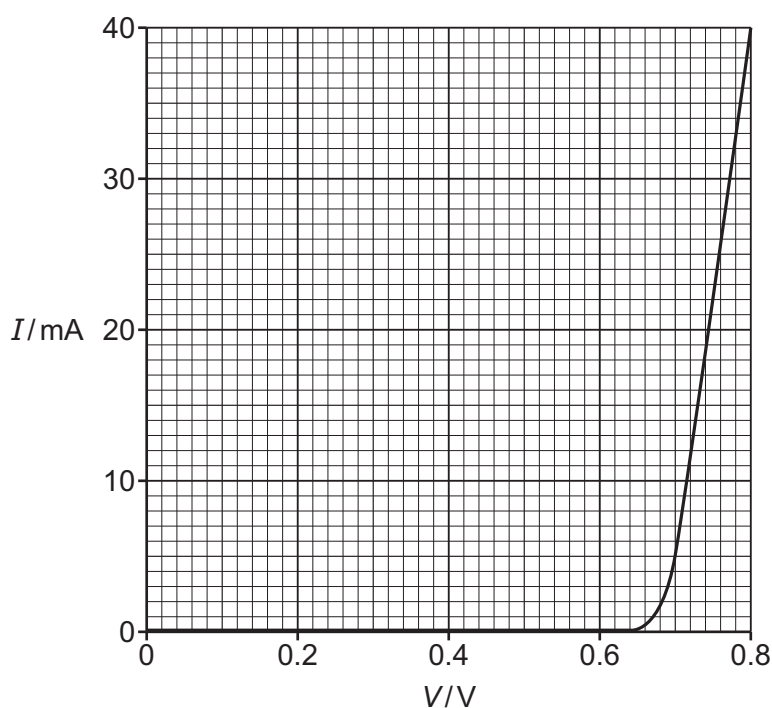


Figure 5.1

- (a) State the name of electrical component X.

[1]

- (b) A metallic conductor has a constant resistance of  $40\ \Omega$ . On Figure 5.1, sketch the  $I$ – $V$  characteristic of this conductor between  $V = 0$  and  $V = 0.80\ \text{V}$ .

[2]

- (c) Component X is connected in a circuit with a resistor of resistance  $20\ \Omega$  as shown in Figure 5.2.

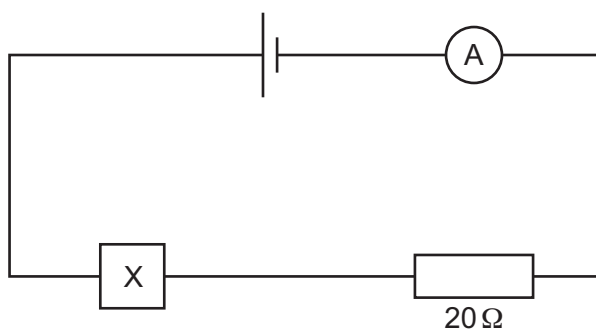


Figure 5.2

The current in the ammeter is  $5.0\ \text{mA}$  and the cell has negligible internal resistance.

Show that the electromotive force (e.m.f.) of the cell is  $0.80\ \text{V}$ .

[3]

- (d) The terminals of the cell in 5(c) are now reversed without making any other changes to the circuit.

State the reading on the ammeter.

reading on ammeter =

A [1]

**4. M/J 26/P24/Q6/a,b,c**

- 6** A nichrome wire of cross-sectional area  $5.7 \times 10^{-8} \text{ m}^2$  is connected to a power supply. The potential difference (p.d.) across the wire is 230 V. The power dissipated in the wire is 1.0 kW.

(a) Show that the resistance of the wire is  $53 \Omega$ .

[2]

(b) The resistivity of nichrome is  $1.3 \times 10^{-6} \Omega \text{ m}$ .

Calculate the length of the wire.

length = \_\_\_\_\_ m [2]

(c) The average drift speed of the charge carriers (free electrons) in the nichrome wire is  $5.4 \times 10^{-3} \text{ m s}^{-1}$ .

Calculate the number density of charge carriers of nichrome.

number density = \_\_\_\_\_  $\text{m}^{-3}$  [3]

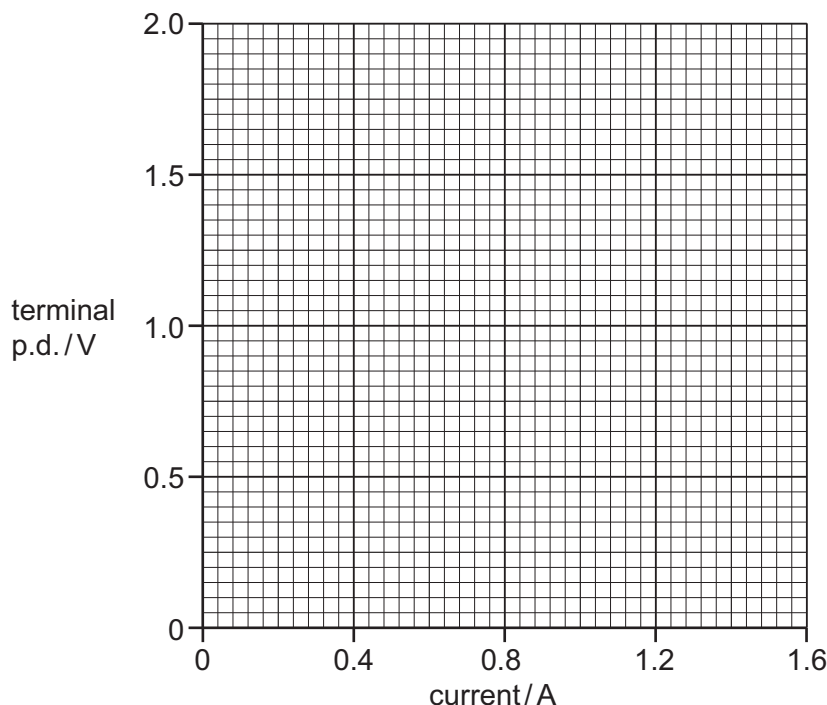
**5. MAR 26/P22/Q6/a,c(i,iii)**

- 6** (a) State what is meant by an electric current.

[1]

(b) A cell has electromotive force (e.m.f.) 1.50 V and internal resistance  $0.75 \Omega$ .

On Figure 6.1, draw the variation of the terminal potential difference (p.d.) with current in the cell for current between 0 A and 1.6 A.

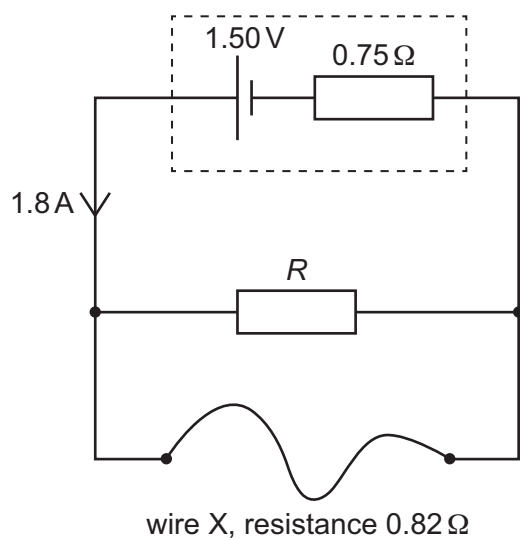


**Figure 6.1**

[2]

- (c) A metal wire X has length 6.8 m and cross-sectional area  $2.0 \times 10^{-7} \text{ m}^2$ . The resistance of the wire is  $0.82 \Omega$ .

The wire is connected to a circuit containing the cell in **6(b)** and a fixed resistor of resistance  $R$ , as shown in Figure 6.2.

**Figure 6.2**

The current in the cell is  $1.8\text{ A}$ .

- (i) Calculate the resistivity  $\rho$  of the metal from which wire X is made.

$$\rho =$$

 $\Omega\text{ m}$  [2]

- (iii) Calculate the power dissipated in the internal resistance of the cell.

$$\text{power} =$$

 $\text{W}$  [2]

**6. O/N 25/P21/Q3/b(i),c(i),d(i)**

- 3** A wire has length  $L$  and cross-sectional area  $A$ . The wire is made from a metal that has Young modulus  $E$  and resistivity  $\rho$ .

- (b) (i)** State an expression, in terms of some or all of  $L$ ,  $A$ ,  $E$  and  $\rho$ , for the resistance  $R_0$  of the wire.

$$R_0 = \quad [1]$$

- (c)** The wire is stretched, within the limit of proportionality, by a tensile force  $F$ . Assume that any changes in the cross-sectional area of the wire are negligible.

- (i)** On Fig. 3.1, sketch the variation with  $F$  of the resistance  $R$  of the wire.



**Fig. 3.1**

- (d)** Copper has a resistivity of  $1.8 \times 10^{-8} \Omega \text{ m}$  and a Young modulus of  $1.3 \times 10^{11} \text{ Pa}$ . [1]

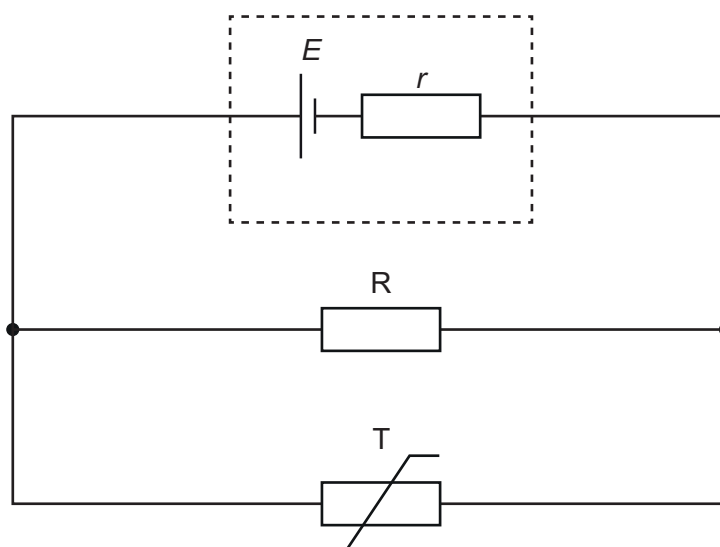
A copper wire of diameter 1.6 mm has a resistance of  $0.034 \Omega$ .

- (i)** Show that the length of the wire is 3.8 m.

[1]

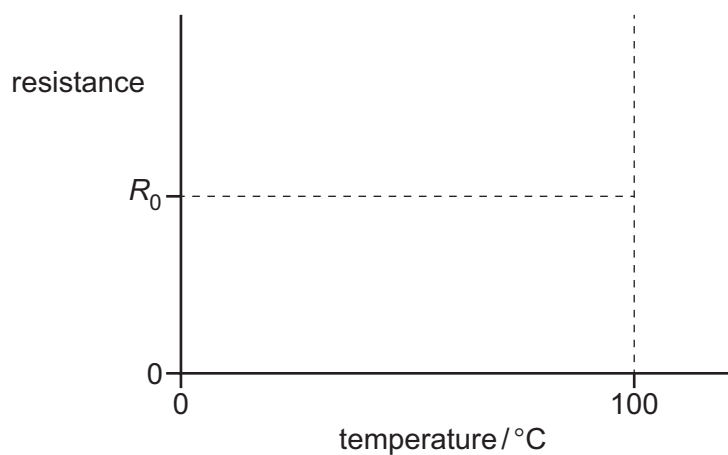
**7. O/N 25/P21/Q5/b(i)**

(b) Fig. 5.1 shows a circuit containing a thermistor  $T$  that has a negative temperature coefficient.

**Fig. 5.1**

- (i) The thermistor has resistance  $R_0$  at a temperature of  $0^\circ\text{C}$ .

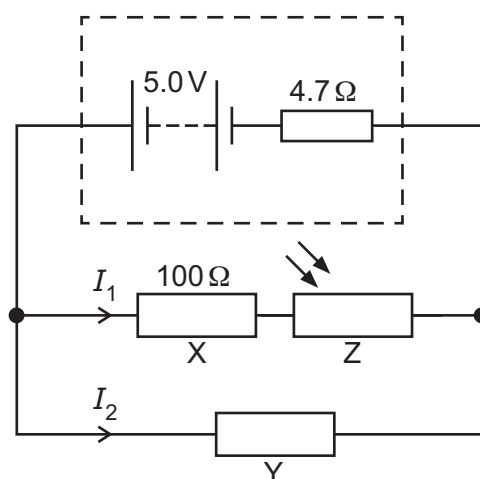
On Fig. 5.2, sketch a possible variation of the resistance of the thermistor with temperature between  $0^\circ\text{C}$  and  $100^\circ\text{C}$ .

**Fig. 5.2**

[2]

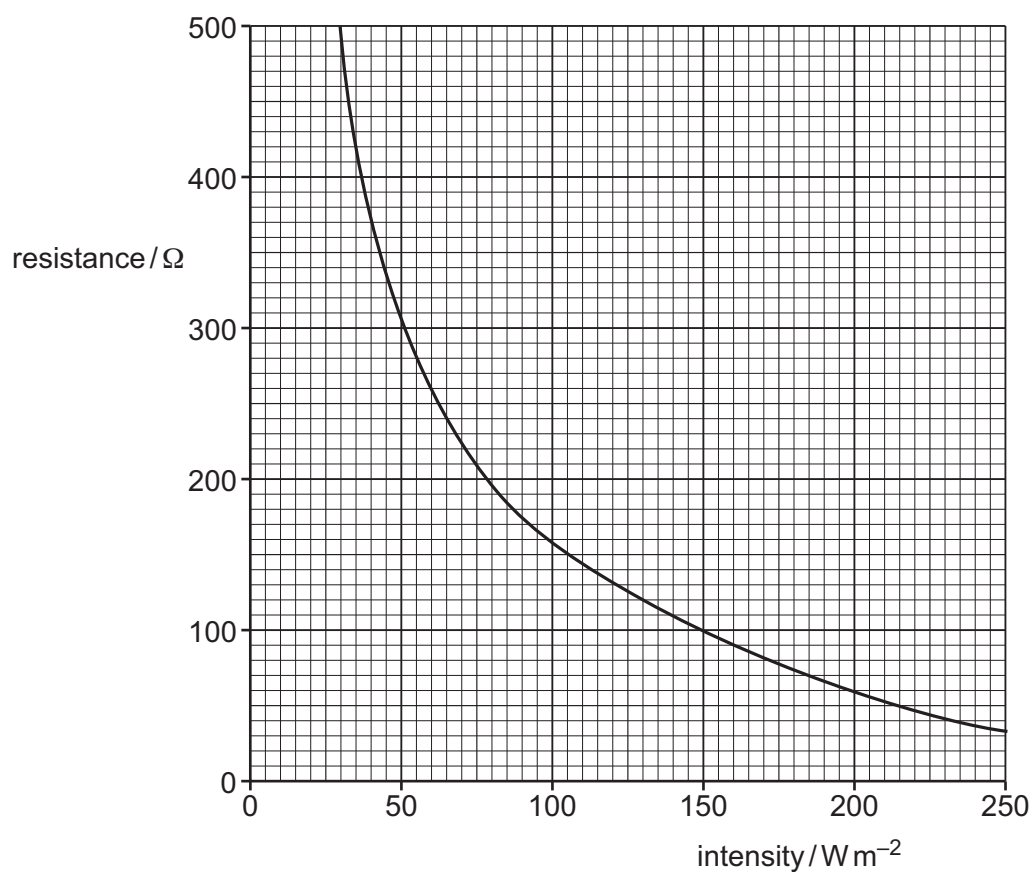
**8. O/N 25/P22/Q5/b(iii)**

- 5 Fig. 5.1 shows a circuit containing a battery, two fixed resistors  $X$  and  $Y$ , and a light-dependent resistor (LDR)  $Z$ .

**Fig. 5.1**

The battery has electromotive force (e.m.f.) 5.0 V and internal resistance  $4.7\ \Omega$ . The current in X is  $I_1$  and the current in Y is  $I_2$ .

The resistance of X is  $100\ \Omega$ . The resistance of Z varies with the intensity of light incident on it as shown in Fig. 5.2.

**Fig. 5.2**

(b) The intensity of light incident on Z is  $130\ \text{W m}^{-2}$ . The current in the battery is 38 mA.

(iii) Calculate the power dissipated in Y.

power =

W [2]

**9. O/N 25/P24/Q5/a(i),b(i,ii)**

- 5** A student uses a circuit containing an ammeter, a voltmeter and a cell to take measurements to determine the resistance of a length of nichrome wire.

**(a) (i)** Define resistance.

[1]

- (b)** The student also measures the length and the diameter of the wire. Table 5.1 shows the measurements recorded for each quantity.

**Table 5.1**

| quantity          | measurement                  |
|-------------------|------------------------------|
| length            | $(0.864 \pm 0.001)\text{m}$  |
| diameter          | $(0.496 \pm 0.002)\text{mm}$ |
| voltmeter reading | $(1.38 \pm 0.02)\text{V}$    |
| ammeter reading   | $(0.276 \pm 0.001)\text{A}$  |

- (i)** Show that the resistance of the wire is  $5.00\,\Omega$ .

[1]

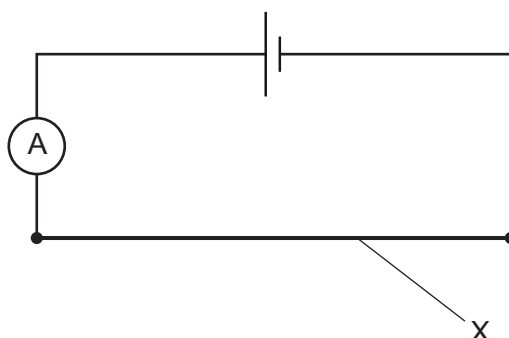
- (ii)** Calculate, to three significant figures, the resistivity  $\rho$  of the nichrome.

$\rho =$

$\Omega\text{m}$  [3]

**10. M/J 25/P21/Q6/a,b(ii)**

- 6** A nichrome wire X of length 45 cm and cross-sectional area  $4.7 \times 10^{-7} \text{ m}^2$  is connected into the circuit shown in Fig. 6.1.

**Fig. 6.1**

The resistance of X is  $1.1 \Omega$ . The cell has electromotive force (e.m.f.)  $1.3 \text{ V}$  and negligible internal resistance.

- (a) (i)** Calculate the current in X.

current = \_\_\_\_\_ A [1]

- (ii)** The number density of charge carriers (electrons) in nichrome is  $8.5 \times 10^{28} \text{ m}^{-3}$ .

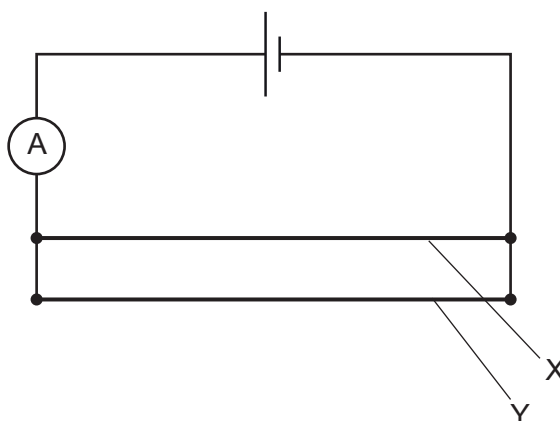
Calculate the average drift speed of the charge carriers in X.

average drift speed = \_\_\_\_\_  $\text{ms}^{-1}$  [2]

- (iii)** Calculate the resistivity of the nichrome.

resistivity = \_\_\_\_\_  $\Omega \text{ m}$  [3]

- (b)** Wire Y is identical to wire X. Wire Y is added to the circuit in parallel with wire X, as shown in Fig. 6.2.

**Fig. 6.2**

State and explain the effect, if any, of this change on:

- (ii)** the average drift speed of the charge carriers in X.

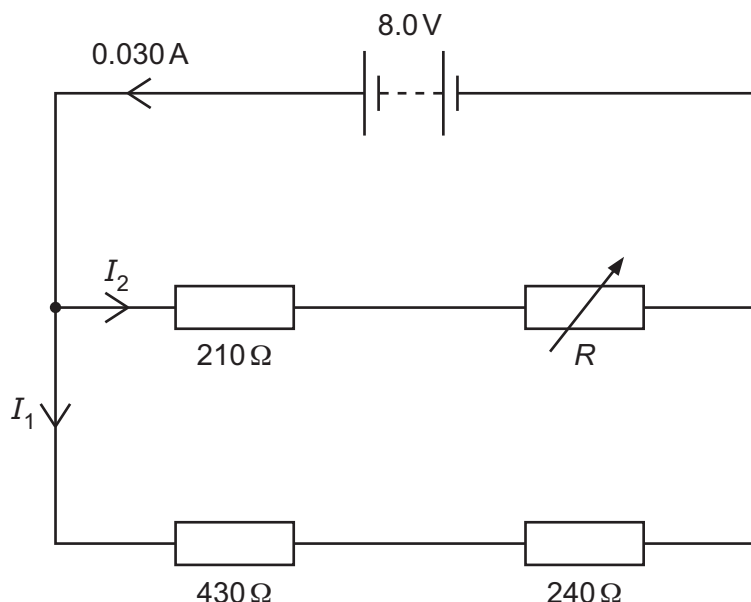
[1]

**11. M/J 25/P22/Q6/a,b(i)**

- 6 (a) Define electric potential difference across a component.

[1]

- (b) A circuit contains four resistors and a battery of electromotive force (e.m.f.)  $8.0\text{ V}$  with negligible internal resistance. When the variable resistor has resistance  $R$ , the currents in the circuit are  $0.030\text{ A}$ ,  $I_1$  and  $I_2$ , as shown in Fig. 6.1.

**Fig. 6.1**

- (i) Determine the charge passing through the battery in a time of  $4.0$  minutes.

charge =

C [2]

**12. M/J 25/P23/Q7/a**

- 7 A nichrome resistance wire has length  $150\text{ cm}$ , cross-sectional area  $2.45 \times 10^{-7}\text{ m}^2$  and resistivity  $1.12 \times 10^{-6}\text{ }\Omega\text{ m}$ .

- (a) Calculate, to three significant figures, the resistance of the wire.

resistance =

 $\Omega$  [3]**13. M/J 25/P24/Q5/a,b**

- 5 (a) (i) State and explain the effect, if any, on the resistance of a filament wire in a lamp as the current in the wire decreases.

[1]

- (ii) On Fig. 5.1, sketch the  $I$ – $V$  characteristic of a filament lamp.

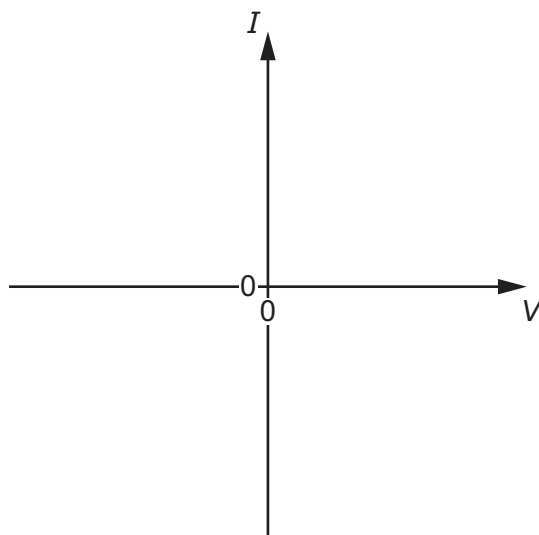


Fig. 5.1

[2]

- (b) A battery of electromotive force (e.m.f.)  $E$  and negligible internal resistance is connected in parallel with two filament lamps A and B, as shown in Fig. 5.2.

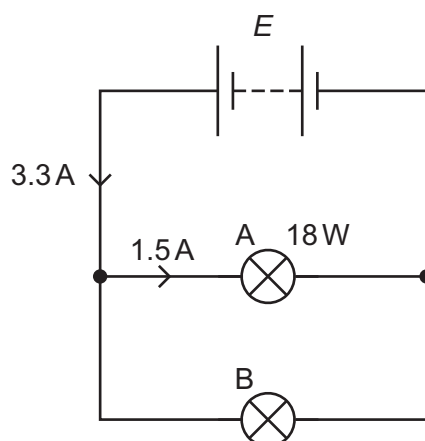


Fig. 5.2

The current in the battery is 3.3 A and the current in lamp A is 1.5 A. The power dissipated in lamp A is 18 W.

- (i) Calculate the e.m.f.  $E$  of the battery.

$$E = \quad \text{V} \quad [2]$$

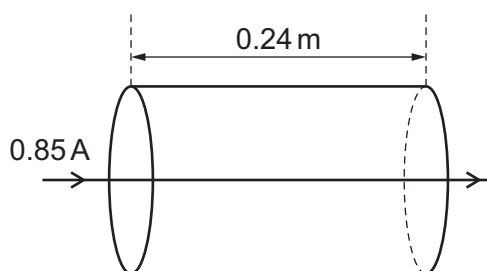
- (ii) The filament wire of lamp B has a cross-sectional area of  $1.4 \times 10^{-9} \text{ m}^2$ . The number of free (conduction) electrons per unit volume in the metal of the filament wire is  $3.4 \times 10^{28} \text{ m}^{-3}$ .

Calculate the average drift speed of the free electrons in the filament wire of lamp B.

$$\text{average drift speed} = \quad \text{ms}^{-1} \quad [3]$$

**14. MAR 25/P22/Q6/a,b,c**

- 6** A cylindrical copper wire P of length 0.24 m is shown in Fig. 6.1.

**Fig. 6.1** (not to scale)

The current in the wire is 0.85 A.

The resistance of the wire is  $3.3 \text{ m}\Omega$ .

The **total** number of charge carriers  $N$  in the wire is  $2.6 \times 10^{22}$ .

The resistivity of copper is  $1.8 \times 10^{-8} \Omega \text{ m}$ .

- (a) Calculate the potential difference between the two ends of the wire.

potential difference = V [2]

- (b) (i) Show that the cross-sectional area of the wire is  $1.3 \times 10^{-6} \text{ m}^2$ .

[2]

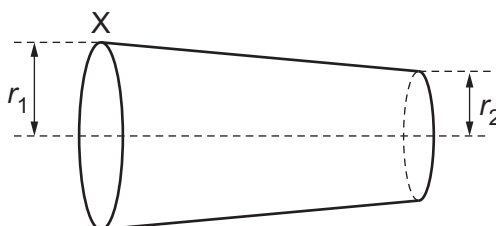
- (ii) Show that the number density of charge carriers in the wire is  $8.3 \times 10^{28} \text{ m}^{-3}$ .

[1]

- (iii) Calculate the average drift speed of the charge carriers (electrons) in the wire.

average drift speed =  $\text{ms}^{-1}$  [2]

- (c) A different copper wire Q has the same volume as wire P, but non-uniform radius, as shown in Fig. 6.2.

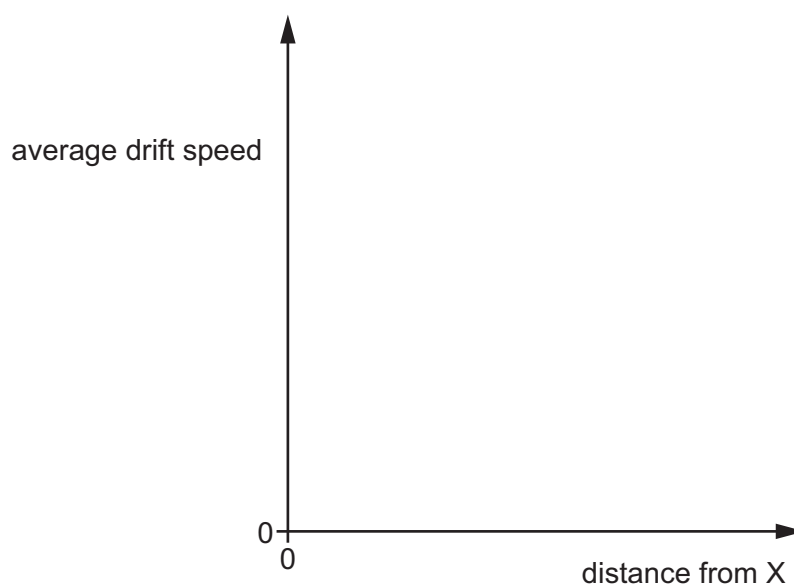
**Fig. 6.2** (not to scale)

The radius  $r_1$  at end X of wire Q is the same as the radius of wire P. Radius  $r_2$  is less than  $r_1$ .

- (i) State and explain how the resistance of wire Q compares with the resistance of wire P.

[4]

- (ii) On Fig. 6.3, sketch a graph of the variation of the average drift speed of the charge carriers with distance from end X of wire Q.

**Fig. 6.3**

[2]

**15. O/N 24/P21/Q3/c(iv)**

(c) An electric motor has an input power of 900 W. The motor takes 1.0 minute to lift a load of weight 240 N at constant speed through a vertical height of 150 m. Resistive forces are negligible.

(iv) Some of the power wasted in the motor is dissipated by the resistance of its coil. This dissipated power is 280 W.

The coil of the motor is made from wire of total length 23 m. The wire has a cross-sectional area of  $2.6 \times 10^{-8} \text{ m}^2$  and is made from metal of resistivity  $1.7 \times 10^{-8} \Omega \text{ m}$ .

Calculate the current in the coil.

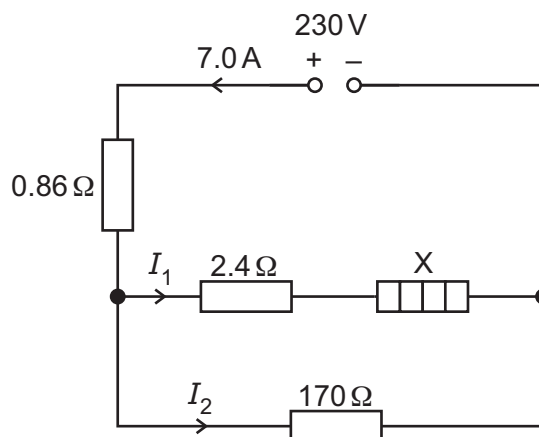
current = A [3]

**16. O/N 24/P22/Q5/a,b(ii,v,vi)**

- 5 (a)** Define electric potential difference (p.d.).

[1]

- (b)** A power supply, three resistors and a component X are connected in the circuit shown in Fig. 5.1.

**Fig. 5.1**

The power supply has an electromotive force (e.m.f.) of 230V and negligible internal resistance. The current in the power supply is 7.0A.

- (ii)** Show that the p.d. across the resistor of resistance  $0.86\Omega$  is 6.0V.

[1]

- (v)** Calculate the power dissipated in component X.

power =

W [2]

- (vi)** The purpose of the circuit is to provide power to component X.

Determine the percentage efficiency of the circuit.

efficiency =

% [2]

**17. O/N 24/P23/Q6/a,b**

**6 (a)** Define resistance.

[1]

**(b)** A cylindrical metal wire of length 2.4 m and cross-sectional area  $8.0 \times 10^{-6} \text{ m}^2$  has a resistance of  $0.33 \Omega$ . There is a current in the wire of 4.7 A.

**(i)** Determine the resistivity of the metal from which the wire is made.

resistivity =  $\Omega \text{ m}$  [2]

**(ii)** Calculate the charge that passes through the wire in a time of 5.0 minutes.

charge = C [2]

**(iii)** The free electrons (charge carriers) in the wire have an average drift speed of  $0.16 \text{ mm s}^{-1}$ .

Determine the number density of charge carriers in the metal.

number density =  $\text{m}^{-3}$  [2]

**18. M/J 24/P22/Q3/a,b,c**

- 3** Lightning occurs when charge builds up in the atmosphere, creating a potential difference between the ground and the atmosphere.

During a lightning strike there is an average current of  $3.3 \times 10^4 \text{ A}$  for a time of  $2.6 \times 10^{-5} \text{ s}$ .

- (a)** Calculate the charge transferred during the lightning strike.

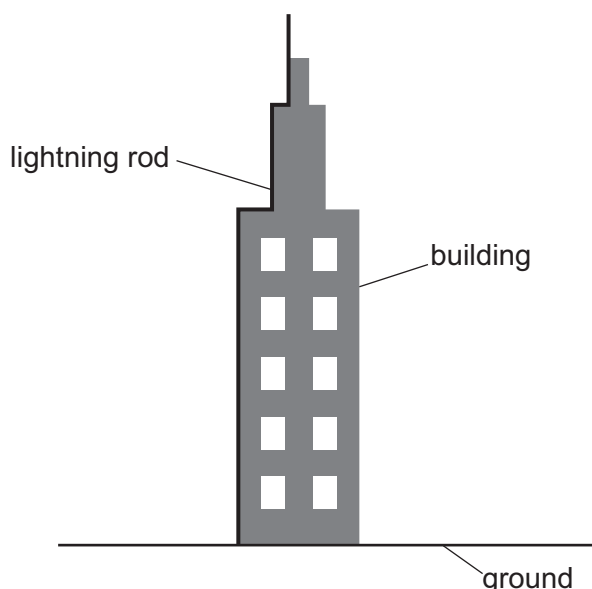
charge = C [2]

- (b)** The potential difference between the ground and the atmosphere is  $3.0 \times 10^7 \text{ V}$ .

Calculate the average power, in GW, transferred during the lightning strike.

power = GW [2]

- (c)** A lightning rod is attached to a tall building to conduct charge safely to the ground. The lightning rod is modelled as a uniform cylindrical copper cable of total length 95 m that runs from the ground to the top of the building, as shown in Fig. 3.1.



**Fig. 3.1**

- (i)** The resistance of the lightning rod is  $9.6 \Omega$ .  
The resistivity of copper is  $1.7 \times 10^{-8} \Omega \text{ m}$ .

Determine the radius of the lightning rod.

radius = m [3]

- (ii)** The radius of the copper lightning rod is doubled with no change to its length.

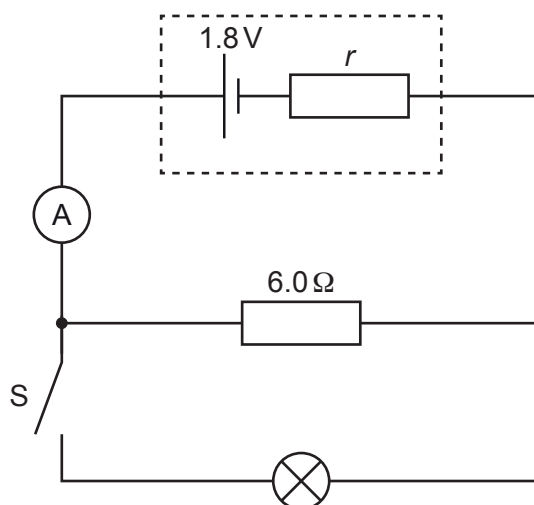
State the effect of this change on the resistance of the lightning rod.

[1]

**19. MAR 24/P22/Q7/a,c(ii)**

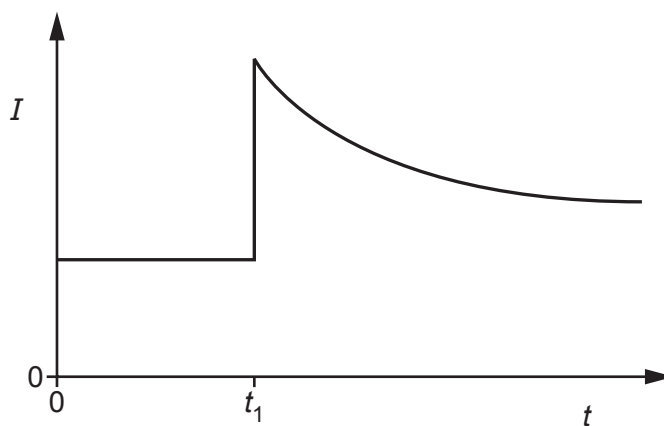
- 7 (a)** Define electric potential difference.

[1]



**Fig. 7.1**

- (c)** At time  $t_1$  switch S in Fig. 7.1 is closed. Fig. 7.2 shows the variation with time  $t$  of the ammeter reading  $I$ .



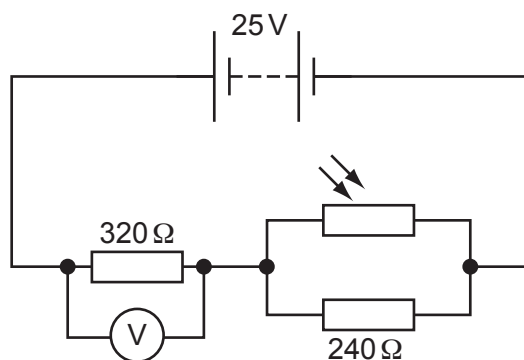
**Fig. 7.2**

- (ii)** By considering the effect of the lamp on the total resistance of the circuit, explain the variation of the ammeter reading shown in Fig. 7.2.

[3]

**20. O/N 23/P21/Q6/a(i,iii)**

- 6** A battery is connected in a circuit with a light-dependent resistor (LDR), two fixed resistors and a voltmeter, as shown in Fig. 6.1.



**Fig. 6.1**

The battery has an electromotive force (e.m.f.) of 25 V and negligible internal resistance. The resistors have resistances of 320  $\Omega$  and 240  $\Omega$ .

- (a)** The voltmeter displays a reading of 16 V.

- (i)** Show that the current in the battery is 0.050 A.

[1]

- (iii)** Determine the ratio

$$\frac{\text{power dissipated in the LDR}}{\text{power dissipated in the } 240\,\Omega \text{ resistor}}.$$

ratio =

[2]

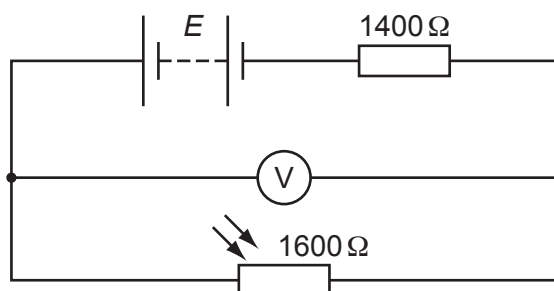
**21. O/N 23/P22/Q6/a,b(i,ii,iv),c(i)**

- 6 (a)** A metal wire has a resistance per unit length of  $0.92 \Omega \text{m}^{-1}$ . The wire has a uniform cross-sectional area of  $5.3 \times 10^{-7} \text{m}^2$ .

Calculate the resistivity of the metal of the wire.

resistivity =  $\Omega \text{m}$  [2]

- (b)** A battery of electromotive force (e.m.f.)  $E$  and negligible internal resistance is connected in series with a fixed resistor and a light-dependent resistor (LDR), as shown in Fig. 6.1.



**Fig. 6.1**

The resistance of the fixed resistor is  $1400 \Omega$ . The intensity of the light illuminating the LDR causes it to have a resistance of  $1600 \Omega$ . A voltmeter connected across the LDR reads  $6.4 \text{V}$ .

- (i)** Show that the current in the LDR is  $4.0 \times 10^{-3} \text{A}$ .

[1]

- (ii)** Calculate the number of free electrons passing through the LDR in a time of 3.2 minutes.

number of free electrons = [2]

- (iv)** Determine the ratio

$$\frac{\text{power dissipated in LDR}}{\text{power dissipated in fixed resistor}}.$$

ratio = [2]

- (c)** The environmental conditions change causing a decrease in the resistance of the LDR in **(b)**. The temperature of the environment does not change.

State whether there is a decrease, increase or no change to:

- (i)** the intensity of the light illuminating the LDR

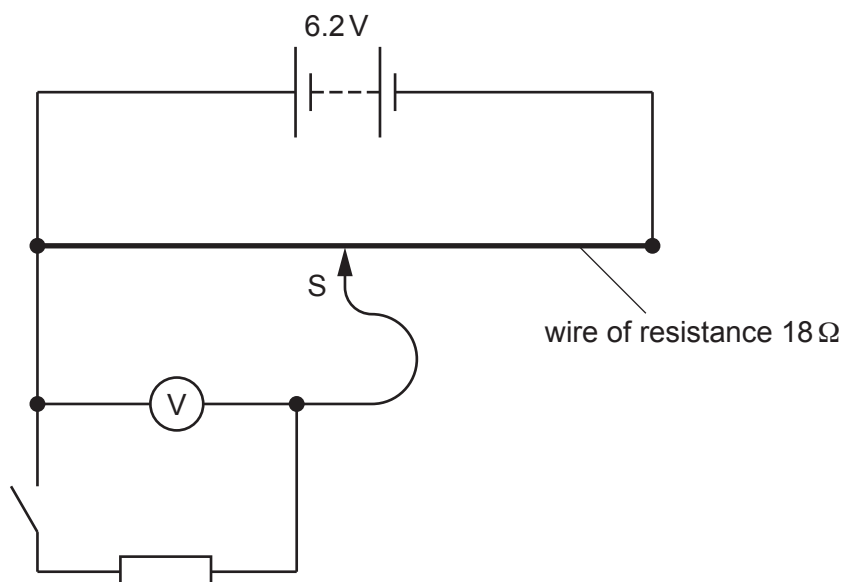
[1]

**22. O/N 23/P23/Q7/a,b(i),c**

7 (a) State Ohm's law.

[2]

- (b) A battery of electromotive force (e.m.f.) 6.2V and negligible internal resistance is connected in a circuit to a uniform resistance wire, a voltmeter, a fixed resistor and a switch, as shown in Fig. 7.1.

**Fig. 7.1**

The resistance wire has resistance  $18\Omega$ , length  $0.94\text{ m}$  and cross-sectional area  $7.2 \times 10^{-8}\text{ m}^2$ . The slider S is positioned half-way along the length of the wire.

- (i) Calculate the resistivity  $\rho$  of the material of the resistance wire.

$\rho =$   $\Omega\text{ m}$  [2]

- (c) The circuit in (b) is altered by changing the battery for one of a different e.m.f. The switch is open.

A student records the following data for the resistance wire:

|                                     |                                      |
|-------------------------------------|--------------------------------------|
| current in the wire                 | 0.93A                                |
| mean drift speed of charge carriers | $1.3 \times 10^{-3}\text{ m s}^{-1}$ |
| number density of charge carriers   | $9.0 \times 10^{28}\text{ m}^{-3}$   |

- (i) Determine the charge  $q$  of a charge carrier in the wire suggested by this data.

$q =$  C [2]

- (ii) With reference to the value of  $q$ , explain why the data recorded by the student cannot be correct.

[1]

**23. M/J 23/P22/Q6**

booklet (scan + OCR)

- (a)**
- The current in a filament lamp decreases.

State and explain how the resistance of the lamp changes.

..... [1]

- (b)**
- A cylindrical wire has length
- $L$
- and resistance
- $R$
- . The
- total**
- number of free electrons (charge carriers) contained in the volume of the wire is
- $N$
- . Each free electron has charge
- $e$
- . The potential difference between the ends of the wire is
- $V$
- .

Determine expressions, in terms of some or all of the symbols  $e$ ,  $L$ ,  $N$ ,  $R$  and  $V$  for:

- (i)**
- the current in the wire

current = ..... [1]

- (ii)**
- the average drift speed of the free electrons

average drift speed = ..... [2]

- (iii)**
- the average time taken for a free electron to move along the full length of the wire.

time taken = ..... [1]

**24. M/J 23/P21/Q6**

booklet (scan + OCR)

A metal wire in a circuit has a length of 1.8 m and a cross-sectional area of  $1.5 \times 10^{-6} \text{ m}^2$ .The total number of free electrons (charge carriers) in the wire is  $2.3 \times 10^{23}$ .

There is a current in the wire so that a charge of 172 C moves past a fixed point in the wire in a time of 2.5 minutes.

- (a)**
- Show that the number density of the free electrons in the wire is
- $8.5 \times 10^{28} \text{ m}^{-3}$
- .

[1]

- (b)**
- Calculate the average drift speed of the free electrons.

average drift speed = .....  $\text{ms}^{-1}$  [3]

**25. M/J 23/P23/Q5**

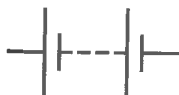
booklet (scan + OCR)

A student sets up a circuit with a battery, an ammeter, a heater and a light-dependent resistor (LDR) all in series.

The battery has negligible internal resistance.

A voltmeter is connected across (in parallel with) the heater.

(a) On Fig. 5.1, complete the circuit diagram of this arrangement.

**Fig. 5.1**

- (b) The heater is a wire made of metal of resistivity  $1.1 \times 10^{-6} \Omega \text{ m}$ . The wire has length 2.0 m and cross-sectional area  $3.8 \times 10^{-7} \text{ m}^2$ . [3]

The reading on the voltmeter is 4.8 V.

Calculate:

- (i) the resistance of the heater

resistance = .....  $\Omega$  [2]

- (ii) the reading on the ammeter,

reading on ammeter = ..... A [1]

- (c) The heater is replaced by a new wire. The new wire is made of the same metal as the wire in (b) and has the same length but a larger diameter.

The resistance of the LDR remains constant.

- (i) State and explain whether the new wire has a resistance that is greater than, less than or the same as that of the wire in (b).

..... [2]

- (ii) State and explain whether the new reading on the voltmeter is greater than, less than or equal to 4.8 V.

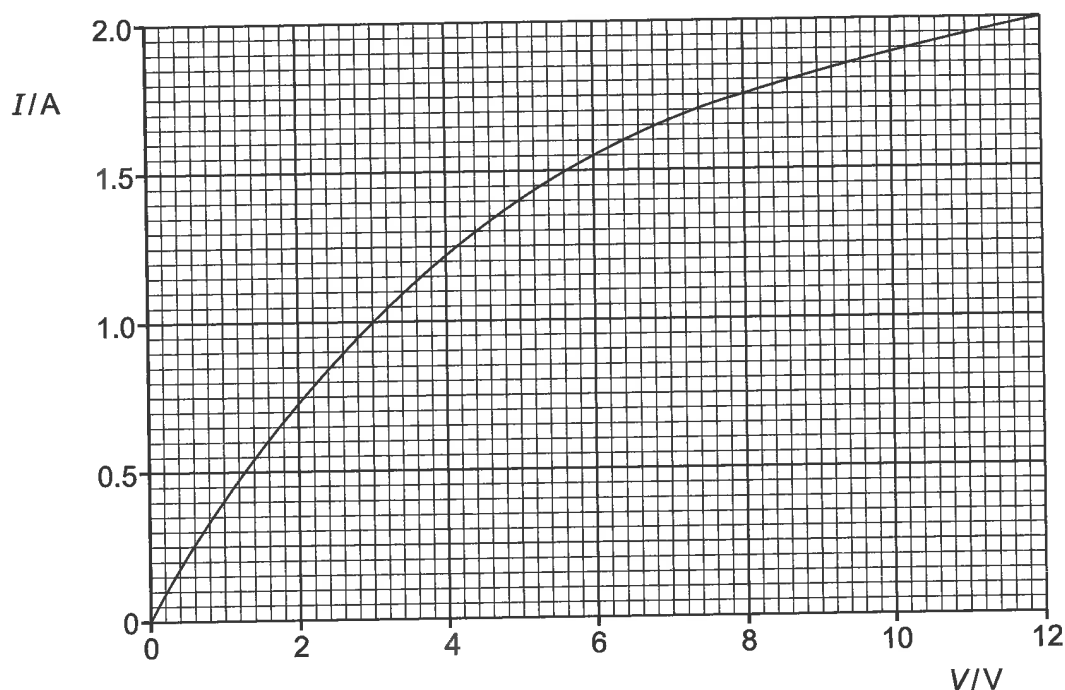
..... [2]

**26. O/N 22/P21/Q5**

booklet (scan + OCR)

**(a)** State Ohm's law.

.....  
..... [2]

**(b)** The variation of current  $I$  with potential difference  $V$  for a filament lamp is shown in Fig. 5.1.**Fig. 5.1**

The resistance of the filament lamp increases with potential difference.

**(i)** State how Fig. 5.1 shows this.

..... [1]

**(ii)** Explain why the resistance varies in this way.

..... [1]

- (c) Fig. 5.2 shows a circuit with a battery of electromotive force (e.m.f.) 12.0V connected to a linear potentiometer AB and two identical filament lamps P and Q.

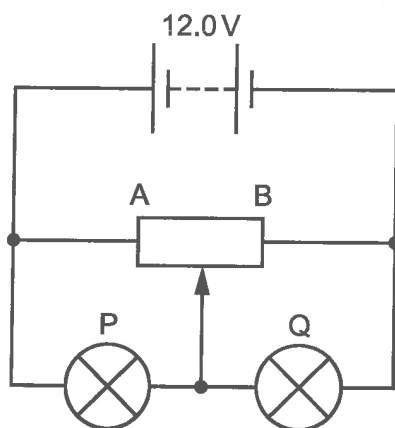


Fig. 5.2

The battery has negligible internal resistance and the lamps each have the same  $I$ - $V$  characteristic shown in Fig. 5.1.

When the slider of the potentiometer is at its midpoint, as shown in Fig. 5.2, the current  $I$  in the battery is 1.78A.

Determine:

- (i) the current in lamp P

current = ..... A[1]

- (ii) the total power dissipated in lamps P and Q

total power = ..... W[2]

- (iii) the resistance of the potentiometer between its ends A and B.

resistance = .....  $\Omega$ [2]

- (d) The slider of the potentiometer in (c) is moved to end A.

State and explain the effect on the brightness of lamps P and Q.

lamp P: .....

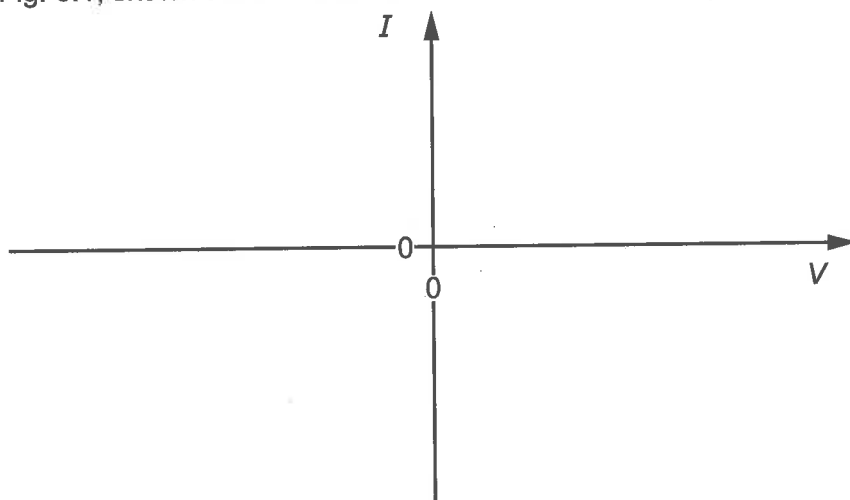
lamp Q: .....

[2]

**27. M/J 22/P23/Q6**

booklet (scan + OCR)

- (a) (i) On Fig. 6.1, sketch the  $I$ – $V$  characteristic of a filament lamp.

**Fig. 6.1**

[2]

- (ii) Explain the shape of the line in (a)(i).

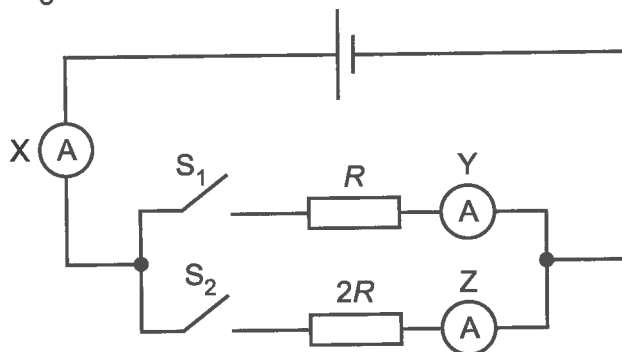
.....  
 .....  
 ..... [3]

- (b) A conducting wire has length  $5.8\text{ m}$  and cross-sectional area  $3.4 \times 10^{-8}\text{ m}^2$ . The resistivity of the metal of the wire is  $5.6 \times 10^{-8}\text{ }\Omega\text{ m}$ .

Calculate the resistance of the wire.

resistance = .....  $\Omega$  [2]

- (c) A resistor of resistance  $R$  is placed in a circuit with a cell of negligible internal resistance, two switches  $S_1$  and  $S_2$ , a second resistor of resistance  $2R$  and three ammeters X, Y and Z. The circuit is shown in Fig. 6.2.

**Fig. 6.2**

The reading on X is 1.0 A when  $S_1$  is open and  $S_2$  is closed.  
Complete Table 6.1.

Table 6.1

| position of switches |        | ammeter readings |                |                |
|----------------------|--------|------------------|----------------|----------------|
| $S_1$                | $S_2$  | reading on X/A   | reading on Y/A | reading on Z/A |
| open                 | open   | 0                | 0              | 0              |
| open                 | closed | 1.0              |                |                |
| closed               | open   |                  |                |                |
| closed               | closed |                  |                |                |

[4]

**28. MAR 22/P22/Q6**

booklet (scan + OCR)

The ends of a metal resistance wire are connected to a battery of electromotive force (e.m.f.) 8.0 V and negligible internal resistance, as shown in Fig. 6.1.

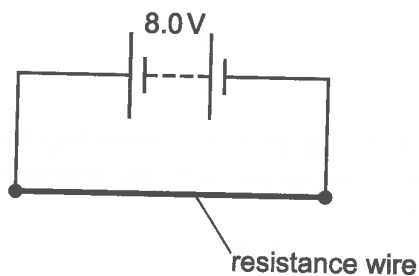


Fig. 6.1

The power dissipated by the resistance wire is 36 W.

(a) Calculate:

(i) the current in the resistance wire

current = ..... A [2]

(ii) the number of free electrons that pass through the resistance wire in a time of 50 s

number = ..... [2]

(iii) the resistance of the wire.

resistance = .....  $\Omega$  [2]

- (b) The metal of the resistance wire in the circuit has a resistivity of  $1.4 \times 10^{-6} \Omega \text{ m}$ . The cross-sectional area of the wire is  $0.25 \text{ mm}^2$ .

Determine the length of the wire.

length = ..... m [2]

- (c) The circuit shown in Fig. 6.1 is modified by replacing the original resistance wire with a second resistance wire. The second wire has a greater diameter than the original wire. There are no other differences between the second wire and the original wire.

By reference to resistance, state and explain whether the power dissipated by the second wire is more than, less than or the same as the power dissipated by the original wire.

..... [2]

- (d) The circuit shown in Fig. 6.1 is modified by connecting a second battery, of e.m.f. 8.0 V and negligible internal resistance, in parallel with the original battery and the original resistance wire, as shown in Fig. 6.2.

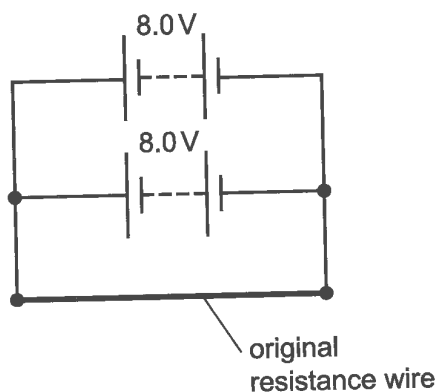


Fig. 6.2

By reference to the current in the resistance wire, state and explain whether the addition of the second battery causes the power in the original resistance wire to decrease, increase or stay the same.

..... [2]

**29. M/J 21/P22/Q5**

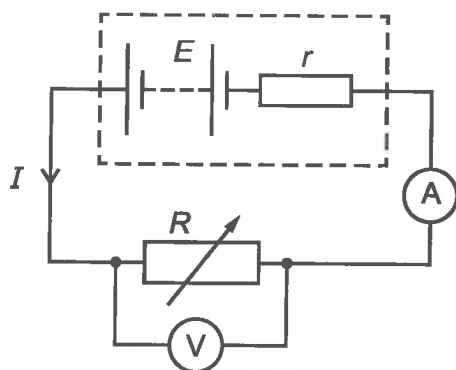
booklet (scan + OCR)

**(a)** Define the *ohm*.

..... [1]

**(b)** A wire is made of metal of resistivity  $\rho$ . The length  $L$  of the wire is gradually increased. Assume that the volume  $V$  of the wire remains constant as its length is increased.Show that the resistance  $R$  of the extending wire is proportional to  $L^2$ .

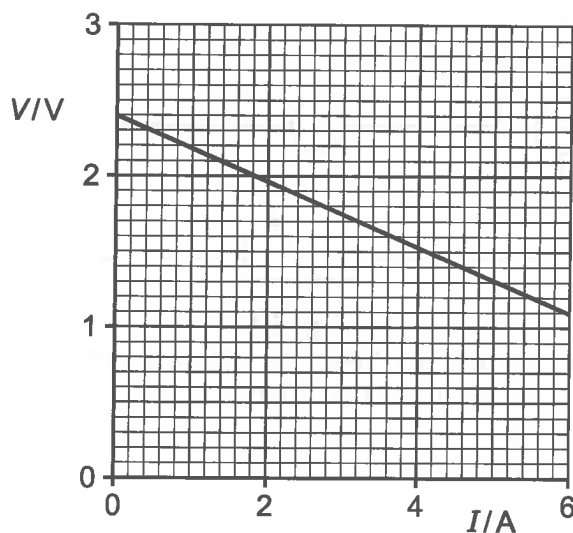
[2]

**(c)** A battery of electromotive force (e.m.f.)  $E$  and internal resistance  $r$  is connected to a variable resistor of resistance  $R$ , as shown in Fig. 5.1.**Fig. 5.1**

An ammeter measures the current  $I$  in the circuit. A voltmeter measures the potential difference  $V$  across the variable resistor.

The resistance  $R$  is now varied to change the values of  $I$  and  $V$ .

The variation with  $I$  of  $V$  is shown in Fig. 5.2.

**Fig. 5.2**

- (i) Use Fig. 5.2 to state the e.m.f.  $E$  of the battery.

$$E = \dots\dots\dots \text{ V [1]}$$

- (ii) Use Fig. 5.2 to determine the power dissipated in the variable resistor when there is a current of 5.0 A.

$$\text{power} = \dots\dots\dots \text{ W [3]}$$

- (iii) State what is represented by the value of the gradient of the graph.

..... [1]

### 30. M/J 21/P23/Q5

booklet (scan + OCR)

- (a) Define the *electromotive force (e.m.f.)* of a source.

..... [2]

- (b) The circuit shown in Fig. 5.1 contains a battery of e.m.f.  $E$  that has internal resistance  $r$ , a variable resistor, a voltmeter and an ammeter.

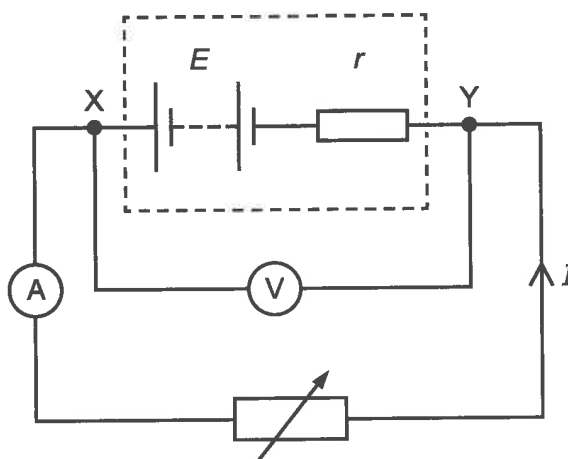


Fig. 5.1

Readings from the two meters are taken for different settings of the variable resistor. The variation with current  $I$  of the potential difference (p.d.)  $V$  across the terminals XY of the battery is shown in Fig. 5.2.

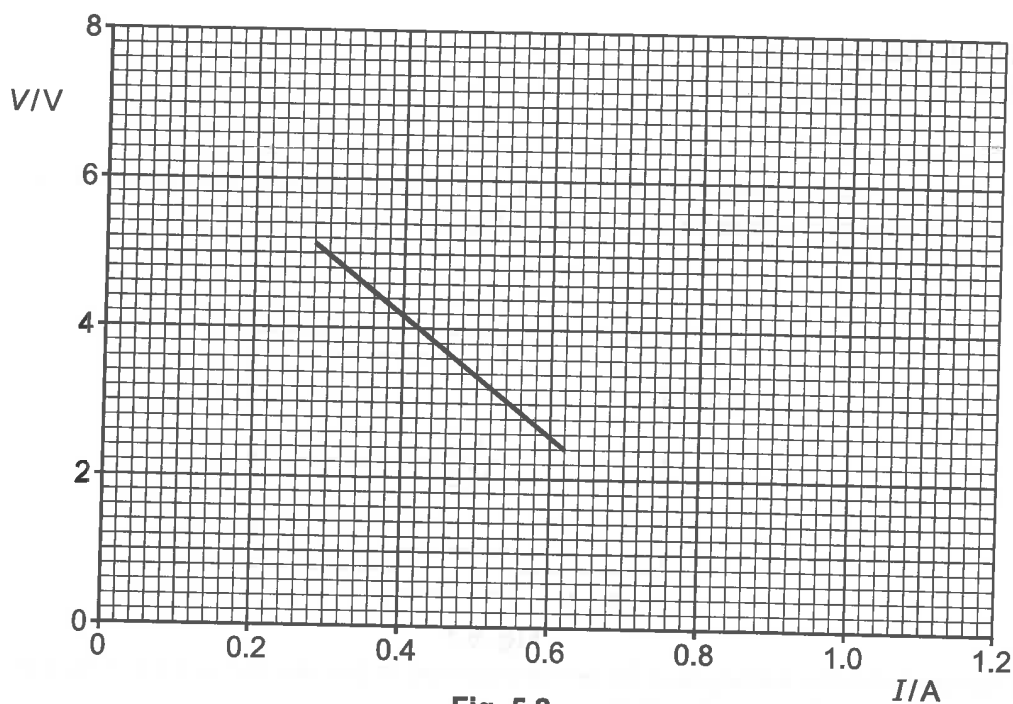


Fig. 5.2

Explain why  $V$  is not constant.

.....

.....

..... [3]

(c) For the battery in (b), use Fig. 5.2 to determine:

(i) the e.m.f.  $E$

$$E = \dots\dots\dots V \quad [1]$$

(ii) the maximum current that the battery can supply

$$\text{maximum current} = \dots\dots\dots A \quad [1]$$

(iii) the internal resistance  $r$ .

$$r = \dots\dots\dots \Omega \quad [2]$$

(d) On Fig. 5.2, sketch a line to show a possible variation with  $I$  of  $V$  for a battery with a lower e.m.f. and a lower internal resistance than the battery in (b). Your line should extend over at least the same range of currents as the original line.

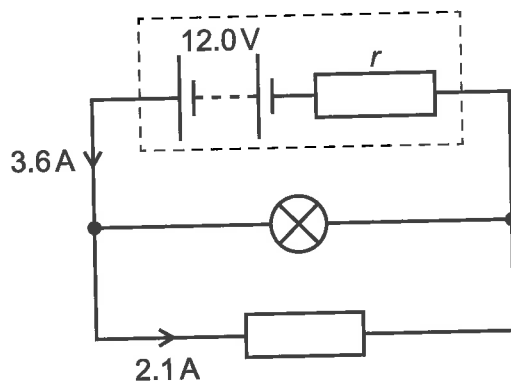
[2]

**31. MAR 21/P22/Q6/a,b(i,ii,iii)**

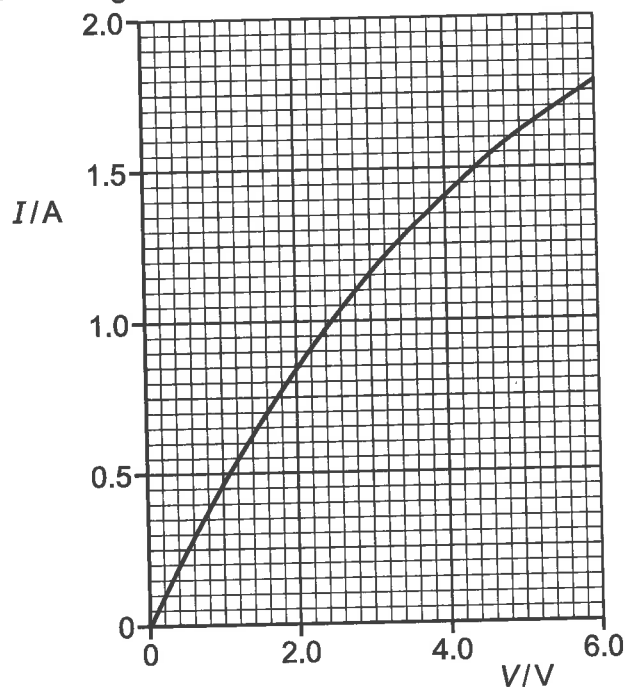
booklet (scan + OCR)

**(a) State Kirchhoff's first law.**

..... [1]

**(b)** A battery of electromotive force (e.m.f.) 12.0V and internal resistance  $r$  is connected to a filament lamp and a resistor, as shown in Fig. 6.1.**Fig. 6.1**

The current in the battery is 3.6A and the current in the resistor is 2.1A. The  $I$ - $V$  characteristic for the lamp is shown in Fig. 6.2.

**Fig. 6.2****(i)** Determine the resistance of the lamp in Fig. 6.1.resistance = .....  $\Omega$  [3]

(ii) Determine the internal resistance  $r$  of the battery.

$$r = \dots\dots\dots \Omega \text{ [2]}$$

(iii) The initial energy stored in the battery is 470 kJ. Assume that the e.m.f. and the current in the battery do not change with time.

Calculate the time taken for the energy stored in the battery to become 240 kJ.

$$\text{time} = \dots\dots\dots \text{ s [2]}$$

**32. O/N 20/P21/Q7**

booklet (scan + OCR)

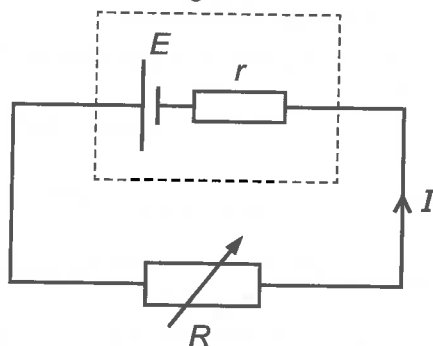
**(a)** Define the *ohm*.

..... [1]  
**(b)** A uniform wire has resistance  $3.2\Omega$ . The wire has length  $2.5\text{m}$  and is made from metal of resistivity  $460\text{n}\Omega\text{m}$ .

Calculate the cross-sectional area of the wire.

cross-sectional area = .....  $\text{m}^2$  [3]

**(c)** A cell of electromotive force (e.m.f.)  $E$  and internal resistance  $r$  is connected to a variable resistor of resistance  $R$ , as shown in Fig. 7.1.

**Fig. 7.1**

The current in the circuit is  $I$ .

**(i)** State, in terms of energy, why the potential difference across the variable resistor is less than the e.m.f. of the cell.

..... [1]  
**(ii)** State an expression for  $E$  in terms of  $I$ ,  $R$  and  $r$ .

$E =$  ..... [1]

**(iii)** The resistance  $R$  of the variable resistor is changed so that it is equal to  $r$ . Determine an expression, in terms of only  $E$  and  $r$ , for the power  $P$  dissipated in the variable resistor.

$P =$  ..... [2]

**33. M/J 20/P21/Q5**

booklet (scan + OCR)

- (a) Metal wire is used to connect a power supply to a lamp. The wire has a total resistance of  $3.4\ \Omega$  and the metal has a resistivity of  $2.6 \times 10^{-8}\ \Omega\text{m}$ . The total length of the wire is  $59\text{ m}$ .
- (i) Show that the wire has a cross-sectional area of  $4.5 \times 10^{-7}\text{ m}^2$ .

[2]

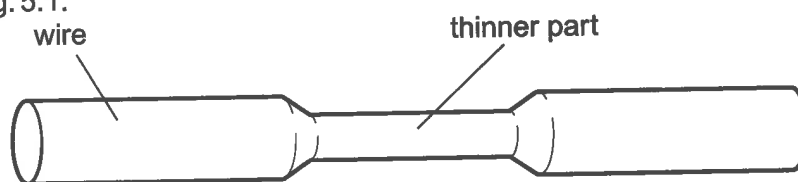
- (ii) The potential difference across the total length of wire is  $1.8\text{ V}$ .  
Calculate the current in the wire.

current = ..... A[1]

- (iii) The number density of the free electrons in the wire is  $6.1 \times 10^{28}\text{ m}^{-3}$ .  
Calculate the average drift speed of the free electrons in the wire.

average drift speed = .....  $\text{ms}^{-1}$  [2]

- (b) A different wire carries a current. This wire has a part that is thinner than the rest of the wire, as shown in Fig. 5.1.

**Fig. 5.1**

- (i) State and explain qualitatively how the average drift speed of the free electrons in the thinner part compares with that in the rest of the wire.

..... [2]

- (ii) State and explain whether the power dissipated in the thinner part is the same, less or more than the power dissipated in an equal length of the rest of the wire.

..... [2]

- (c) Three resistors have resistances of  $180\ \Omega$ ,  $90\ \Omega$  and  $30\ \Omega$ .
- (i) Sketch a diagram showing how **two** of these three resistors may be connected together to give a combined resistance of  $60\ \Omega$  between the terminals shown. Ensure you label the values of the resistances in your diagram.



[1]

- (ii) A potential divider circuit is produced by connecting the three resistors to a battery of electromotive force (e.m.f.)  $12\text{ V}$  and negligible internal resistance. The potential divider circuit provides an output potential difference  $V_{\text{OUT}}$  of  $8.0\text{ V}$ . Fig. 5.2 shows the circuit diagram.

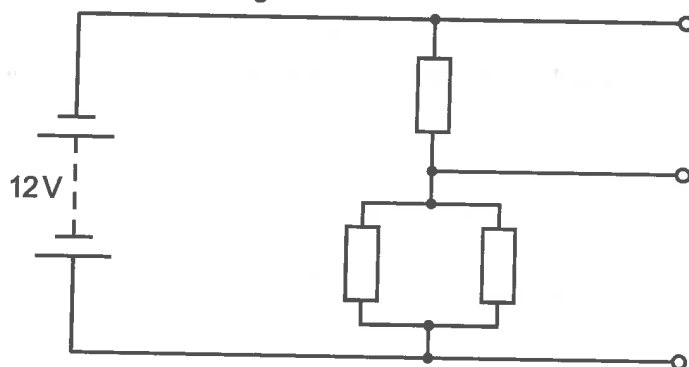


Fig. 5.2

On Fig. 5.2, label the resistances of all three resistors and the potential difference  $V_{\text{OUT}}$ .

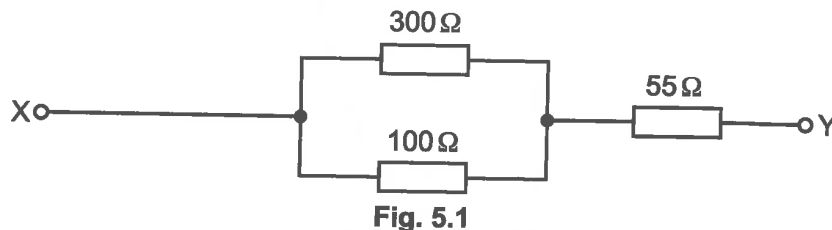
[2]

**34. M/J 20/P23/Q5**

booklet (scan + OCR)

**(a) Define the volt.**

..... [1]

**(b) Fig. 5.1 shows a network of three resistors.**

Calculate:

**(i) the combined resistance of the two resistors connected in parallel**combined resistance = .....  $\Omega$  [1]**(ii) the total resistance between terminals X and Y.**total resistance = .....  $\Omega$  [1]**(c) The network in (b) is connected to a power supply so that there is a potential difference between terminals X and Y. The power dissipated in the resistor of resistance  $55\Omega$  is  $0.20\text{ W}$ .****(i) Calculate the current in the resistor of resistance:**1.  $55\Omega$ 

current = ..... A

2.  $300\Omega$ .

current = ..... A [3]

**(ii) Calculate the potential difference between X and Y.**

potential difference = ..... V [1]

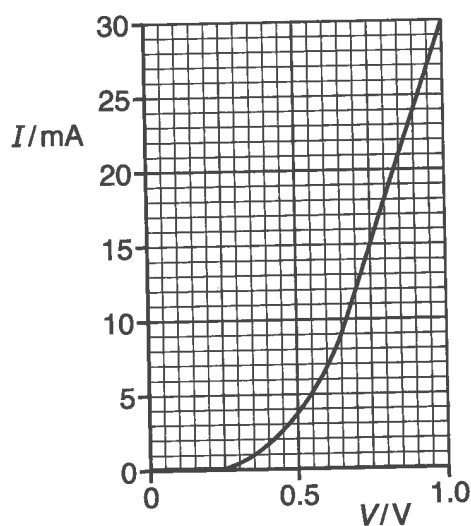
**35. O/N 19/P21/Q6**

booklet (scan + OCR)

**(a)** Define *electric potential difference* (*p.d.*).

[1]

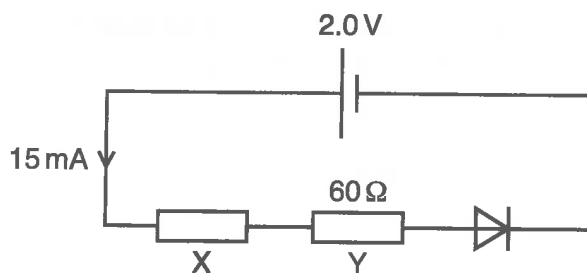
**(b)** The variation with potential difference  $V$  of the current  $I$  in a semiconductor diode is shown in Fig. 6.1.

**Fig. 6.1**

Use Fig. 6.1 to describe qualitatively the variation of the resistance of the diode as  $V$  increases from 0 to 1.0 V.

[2]

**(c)** The diode in **(b)** is part of the circuit shown in Fig. 6.2.

**Fig. 6.2**

The cell of electromotive force (e.m.f.) 2.0 V and negligible internal resistance is connected in series with the diode and resistors X and Y. The resistance of Y is  $60\ \Omega$ . The current in the cell is 15 mA.

(i) Use Fig. 6.1 to determine the resistance of the diode.

resistance = .....  $\Omega$  [3]

(ii) Calculate:

1. the resistance of X

resistance = .....  $\Omega$  [3]

2. the ratio

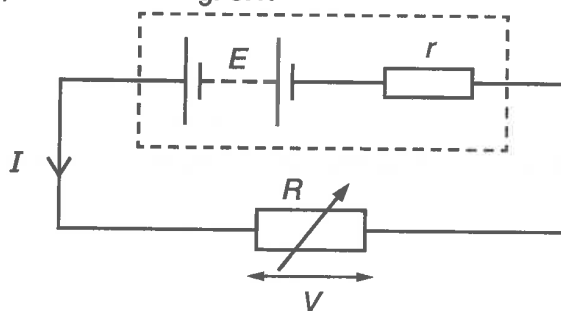
$$\frac{\text{power dissipated in resistor Y}}{\text{total power produced by the cell}}$$

ratio = ..... [2]

### 36. M/J 19/P21/Q6

booklet (scan + OCR)

A battery of electromotive force (e.m.f.)  $E$  and internal resistance  $r$  is connected to a variable resistor of resistance  $R$ , as shown in Fig. 6.1.

**Fig. 6.1**

The current in the circuit is  $I$  and the potential difference across the variable resistor is  $V$ .

- (a) Explain, in terms of energy, why  $V$  is less than  $E$ .

.....[1]

- (b) State an equation relating  $E$ ,  $I$ ,  $r$  and  $V$ .

.....[1]

- (c) The resistance  $R$  of the variable resistor is varied. The variation with  $I$  of  $V$  is shown in Fig. 6.2.

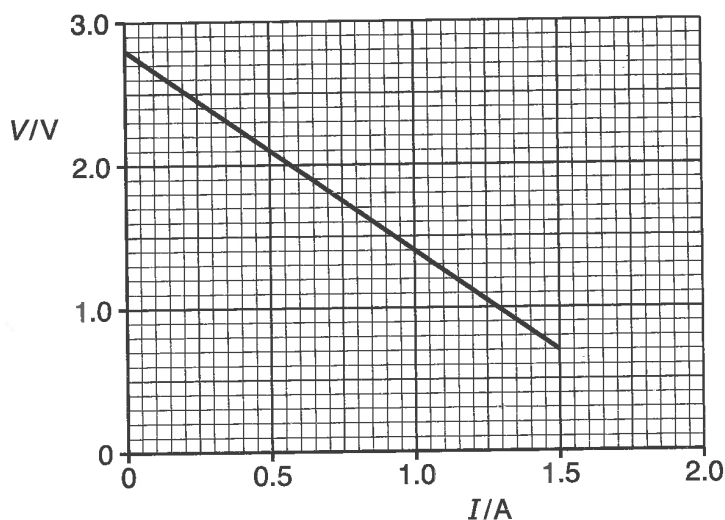


Fig. 6.2

Use Fig. 6.2 to:

- (i) explain how it may be deduced that the e.m.f. of the battery is 2.8 V

.....[1]

- (ii) calculate the internal resistance  $r$ .

$$r = \dots\dots\dots \Omega \text{ [2]}$$

- (d) The battery stores 9.2 kJ of energy. The variable resistor is adjusted so that  $V = 2.1 \text{ V}$ . Use Fig. 6.2 to:

- (i) calculate resistance  $R$

$$R = \dots\dots\dots \Omega \text{ [1]}$$

- (ii) calculate the number of conduction electrons moving through the battery in a time of 1.0 s

$$\text{number} = \dots\dots\dots \text{ [1]}$$

- (iii) determine the time taken for the energy in the battery to become equal to 1.6 kJ.  
(Assume that the e.m.f. of the battery and the current in the battery remain constant.)

time taken = ..... s [3]

### 37. MAR 19/P22/Q6

booklet (scan + OCR)

- (a) Using energy transformations, describe the *electromotive force* (e.m.f.) of a battery and the *potential difference* (p.d.) across a resistor.

e.m.f.: .....

p.d.: ..... [2]

- (b) A battery of e.m.f. 6.0 V and negligible internal resistance is connected to a network of resistors and a voltmeter, as shown in Fig. 6.1.

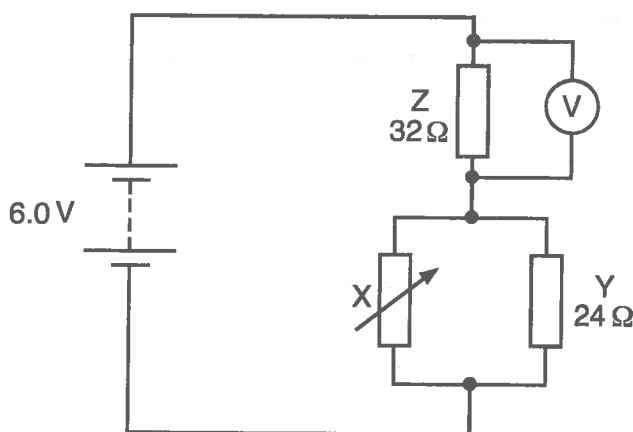


Fig. 6.1

Resistor Y has a resistance of 24 Ω and resistor Z has a resistance of 32 Ω.

- (i) The resistance  $R_X$  of the variable resistor X is adjusted until the voltmeter reads 4.8 V.  
Calculate:

1. the current in resistor Z

current = ..... A [1]

2. the total power provided by the battery

power = ..... W [2]

3. the number of conduction electrons that move through the battery in a time interval

of 25 s

number = ..... [2]

4. the total resistance of X and Y connected in parallel

total resistance = .....  $\Omega$  [2]

5. the resistance  $R_X$ .

$R_X$  = .....  $\Omega$  [2]

- (ii) The resistance  $R_X$  is now decreased.

State and explain the change, if any, to the reading on the voltmeter.

.....

..... [2]

**38. O/N 18/P22/Q7**

booklet (scan + OCR)

- (a)**
- The current
- $I$
- in a metal wire is given by the expression

$$I = Anve.$$

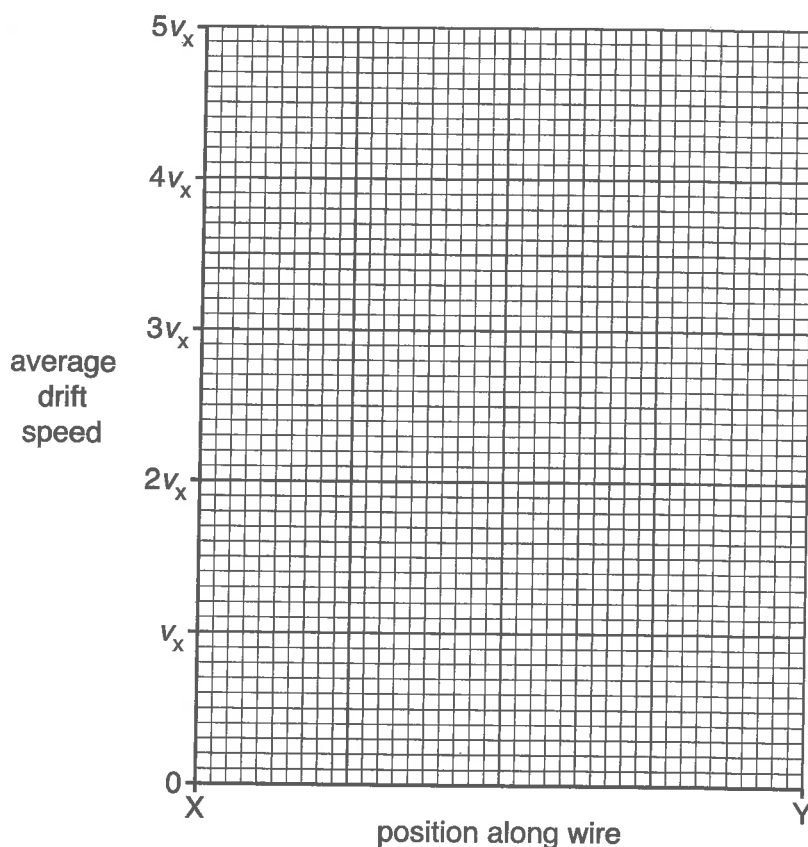
State what is meant by the symbols  $A$  and  $n$ . $A$ : ..... $n$ : ..... [2]

- (b)**
- The diameter of a wire XY varies linearly with distance along the wire as shown in Fig. 7.1.

**Fig. 7.1**

There is a current  $I$  in the wire. At end X of the wire, the diameter is  $d$  and the average drift speed of the free electrons is  $v_x$ . At end Y of the wire, the diameter is  $\frac{d}{2}$ .

On Fig. 7.2, sketch a graph to show the variation of the average drift speed with position along the wire between X and Y.

**Fig. 7.2**

[2]

**39. O/N 18/P23/Q6**

booklet (scan + OCR)

**(a)** Define the *coulomb*.

..... [1]

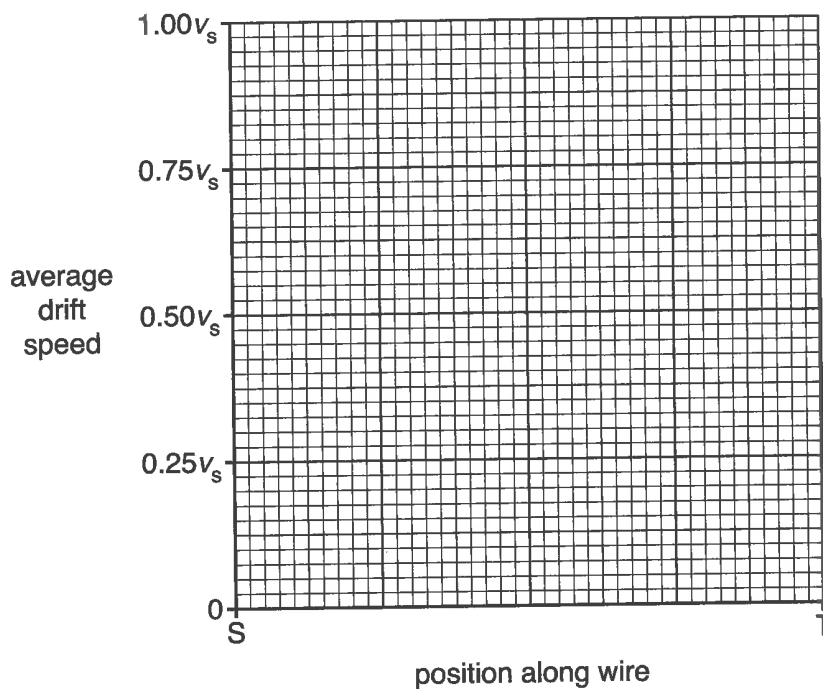
**(b)** An electric current is a flow of charge carriers.

In the following list, underline the possible charges for a charge carrier.

 $8.0 \times 10^{-19} \text{ C}$       $4.0 \times 10^{-19} \text{ C}$       $1.6 \times 10^{-19} \text{ C}$       $1.6 \times 10^{-20} \text{ C}$  [1]
**(c)** The diameter of a wire ST varies linearly with distance along the wire as shown in Fig. 6.1.**Fig. 6.1**

There is a current  $I$  in the wire. At end S of the wire, the diameter is  $d$  and the average drift speed of the free electrons is  $v_s$ . At end T of the wire, the diameter is  $2d$ .

On Fig. 6.2, sketch a graph to show the variation of the average drift speed with position along the wire between S and T.

**Fig. 6.2**

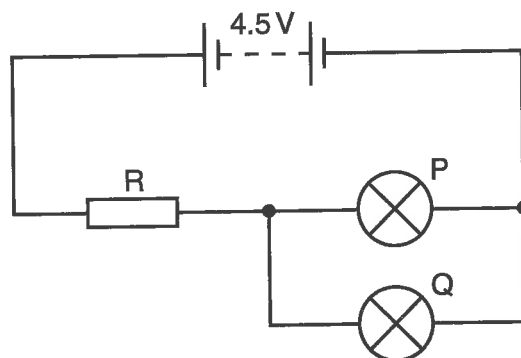
[2]

**40. M/J 18/P21/Q6**

booklet (scan + OCR)

**(a)** Define the *volt*.

..... [1]

**(b)** A battery of electromotive force (e.m.f.) 4.5 V and negligible internal resistance is connected to two filament lamps P and Q and a resistor R, as shown in Fig. 6.1.**Fig. 6.1**

The current in lamp P is 0.15 A.

The  $I$ – $V$  characteristics of the filament lamps are shown in Fig. 6.2.

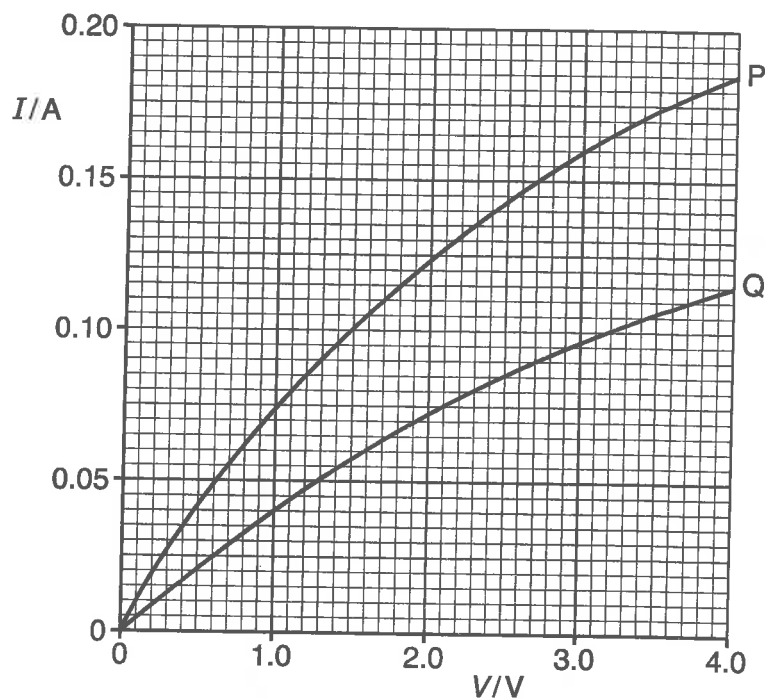


Fig. 6.2

- (i) Use Fig. 6.2 to determine the current in the battery. Explain your working.

current = ..... A [2]

- (ii) Calculate the resistance of resistor R.

resistance = .....  $\Omega$  [2]

- (iii) The filament wires of the two lamps are made from material with the same resistivity at their operating temperature in the circuit. The diameter of the wire of lamp P is twice the diameter of the wire of lamp Q. Determine the ratio

$$\frac{\text{length of filament wire of lamp P}}{\text{length of filament wire of lamp Q}}$$

ratio = ..... [3]

- (iv) The filament wire of lamp Q breaks and stops conducting.

State and explain, qualitatively, the effect on the resistance of lamp P.

.....

..... [2]

**41. O/N 17/P22/Q5**

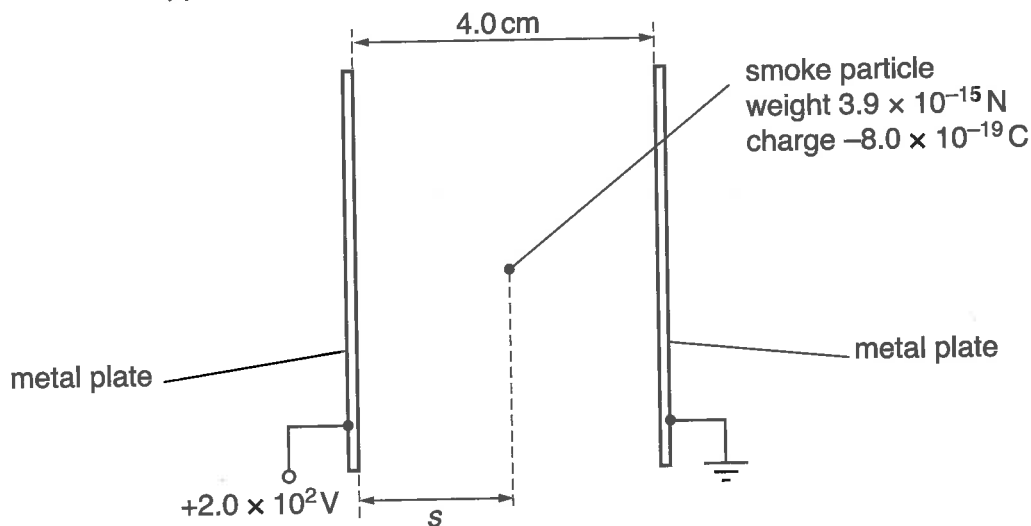
booklet (scan + OCR)

May be outside the 2025–27 syllabus

**(a)** Define the *coulomb*.

[1]

- (b)** Two vertical metal plates in a vacuum have a separation of 4.0 cm. A potential difference of  $2.0 \times 10^2 \text{ V}$  is applied between the plates. Fig. 5.1 shows a side view of this arrangement.

**Fig. 5.1**

A smoke particle is in the uniform electric field between the plates. The particle has weight  $3.9 \times 10^{-15} \text{ N}$  and charge  $-8.0 \times 10^{-19} \text{ C}$ .

- (i)** Show that the electric force acting on the particle is  $4.0 \times 10^{-15} \text{ N}$ .

[2]

- (ii)** On Fig. 5.1, draw labelled arrows to show the directions of the two forces acting on the smoke particle. [1]

- (iii)** The resultant force acting on the particle is  $F$ . Determine

1. the magnitude of  $F$ ,

magnitude = ..... N

2. the angle of  $F$  to the horizontal.

angle = ..... ° [3]

- (c)** The electric field in **(b)** is switched on at time  $t = 0$  when the particle is at a horizontal displacement  $s = 2.0 \text{ cm}$  from the left-hand plate. At time  $t = 0$  the horizontal velocity of the particle is zero. The particle is then moved by the electric field until it hits a plate at time  $t = T$ .

On Fig. 5.2, sketch the variation with time  $t$  of the horizontal displacement  $s$  of the particle from the left-hand plate.

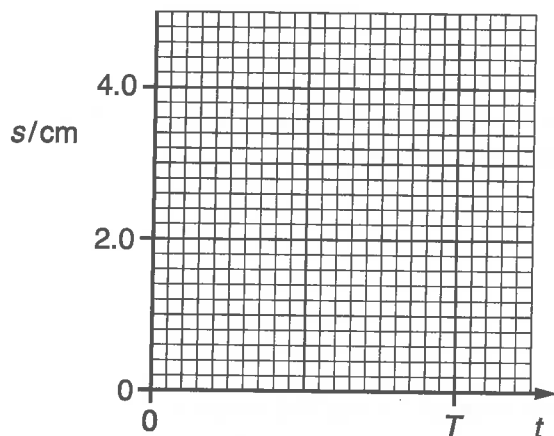


Fig. 5.2

[2]

#### 42. O/N 17/P21/Q7

booklet (scan + OCR)

(a) Define the *ohm*.

[1]

(b) Wires are used to connect a battery of negligible internal resistance to a lamp, as shown in Fig. 7.1.

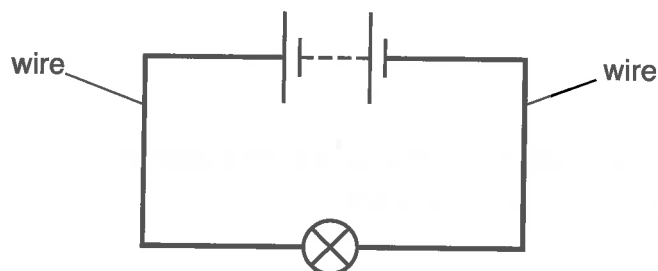


Fig. 7.1

The lamp is at its normal operating temperature. Some data for the filament wire of the lamp and for the connecting wires of the circuit are shown in Fig. 7.2.

|                                                           | filament wire | connecting wires |
|-----------------------------------------------------------|---------------|------------------|
| diameter                                                  | $d$           | $14d$            |
| total length                                              | $L$           | $7.0L$           |
| resistivity of metal<br>(at normal operating temperature) | $\rho$        | $0.028\rho$      |

Fig. 7.2

(i) Show that

$$\frac{\text{resistance of filament wire}}{\text{total resistance of connecting wires}} = 1000.$$

[2]

- (ii) Use the information in (i) to explain qualitatively why the power dissipated in the filament wire of the lamp is greater than the total power dissipated in the connecting wires.

..... [1]

- (iii) The lamp is rated as 12V, 6.0W. Use the information in (i) to determine the total resistance of the connecting wires.

total resistance of connecting wires = .....  $\Omega$  [3]

- (iv) The diameter of the connecting wires is decreased. The total length of the connecting wires and the resistivity of the metal of the connecting wires remain the same.

State and explain the change, if any, that occurs to the resistance of the filament wire of the lamp.

.....

.....

..... [3]

**43. O/N 17/P23/Q6**

booklet (scan + OCR)

A filament lamp is rated as 30 W, 120 V. A potential difference of 120 V is applied across the lamp.

(a) For the filament wire of the lamp, calculate

(i) the current,

current = ..... A [2]

(ii) the number of electrons passing a point in 3.0 hours.

number = ..... [2]

(b) Show that the resistance of the filament wire is  $480\ \Omega$ .

[2]

(c) The filament wire has an uncoiled length of 580 mm and is made of metal. The metal has resistivity  $6.1 \times 10^{-7}\ \Omega\text{ m}$  at the operating temperature of the lamp.

Calculate the diameter of the wire.

diameter = ..... m [3]

(d) The potential difference across the lamp is now reduced. State and explain the effect, if any, on the resistance of the filament wire.

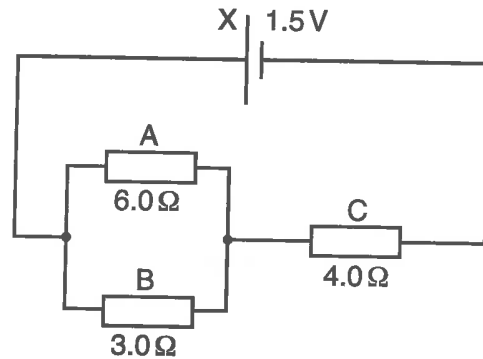
..... [11]

**44. M/J 17/P21/Q6**

booklet (scan + OCR)

**(a)** Define the *ohm*.

..... [1]

**(b)** A cell X of electromotive force (e.m.f.) 1.5 V and negligible internal resistance is connected in series to three resistors A, B and C, as shown in Fig. 6.1.**Fig. 6.1**

Resistors A and B have resistances  $6.0\ \Omega$  and  $3.0\ \Omega$  respectively and are connected in parallel. Resistor C has resistance  $4.0\ \Omega$  and is connected in series with the parallel combination. Calculate

**(i)** the current in the circuit,

current = .....A [3]

**(ii)** the current in resistor B,

current = .....A [1]

**(iii)** the ratio

$$\frac{\text{power dissipated in resistor B}}{\text{power dissipated in resistor C}}$$

ratio = ..... [2]

- (c) The resistors A, B and C in (b) are wires of the same material and have the same length.

(i) Explain how the resistors may be made with different resistance values.

.....[1]

(ii) Calculate the ratio

$$\frac{\text{average drift speed of the charge carriers in resistor B}}{\text{average drift speed of the charge carriers in resistor C}}$$

ratio = .....[2]

- (d) A cell of e.m.f. 1.5 V and negligible internal resistance is connected in parallel with cell X in Fig. 6.1 with their positive terminals together.

State the change, if any, to the current in

(i) cell X,

.....[1]

(ii) resistor C.

.....[1]

#### 45. M/J 17/P23/Q6

booklet (scan + OCR)

- (a) Describe the  $I$ – $V$  characteristic of

(i) a metallic conductor at constant temperature,

.....[1]

(ii) a semiconductor diode.

.....[2]

- (b) Two identical filament lamps are connected in series and then in parallel to a battery of electromotive force (e.m.f.) 12 V and negligible internal resistance, as shown in Fig. 6.1a and Fig. 6.1b.

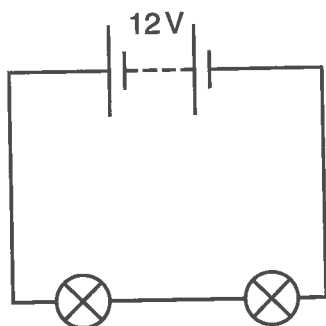


Fig. 6.1a

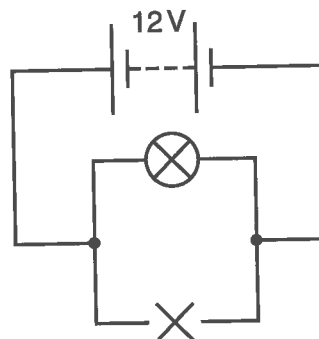


Fig. 6.1b

The  $I$ – $V$  characteristic of each lamp is shown in Fig. 6.2.

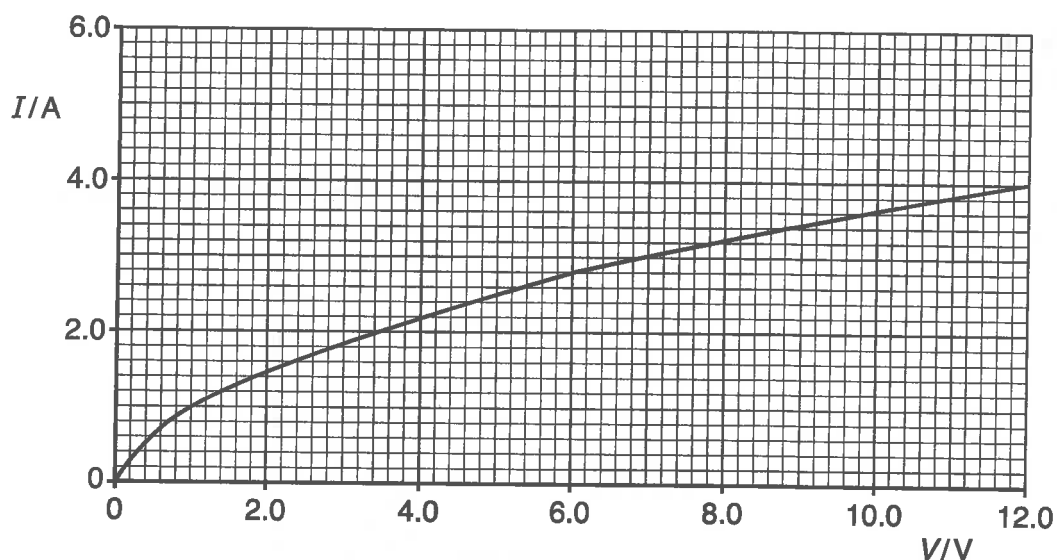


Fig. 6.2

- (i) Use the information shown in Fig. 6.2 to determine the current through the battery in
- the circuit of Fig. 6.1a,

current = ..... A

- the circuit of Fig. 6.1b.

current = ..... A [3]

- (ii) Calculate the total resistance in

- the circuit of Fig. 6.1a,

resistance = .....  $\Omega$

- the circuit of Fig. 6.1b.

resistance = .....  $\Omega$  [3]

- (iii) Calculate the ratio

$\frac{\text{power dissipated in a lamp in the circuit of Fig. 6.1a}}{\text{power dissipated in a lamp in the circuit of Fig. 6.1b}}$

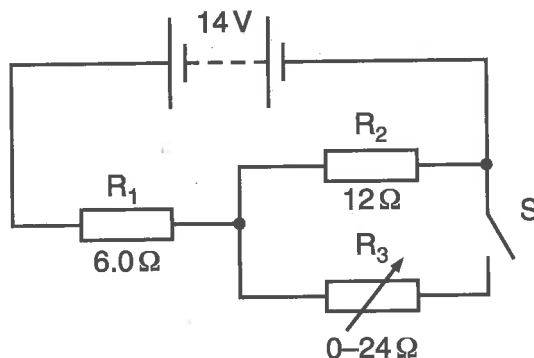
ratio = ..... [2]

**46. O/N 16/P21/Q6**

booklet (scan + OCR)

**(a)** Define electric *potential difference* (*p.d.*).

..... [1]

**(b)** A battery of electromotive force (e.m.f.) 14 V and negligible internal resistance is connected to a resistor network, as shown in Fig. 6.1.**Fig. 6.1**

$R_1$  and  $R_2$  are fixed resistors of resistances  $6.0\Omega$  and  $12\Omega$  respectively.  $R_3$  is a variable resistor.

Switch S is **closed**.

**(i)** Calculate the current in the battery when the resistance of  $R_3$  is set

1. at zero,

current = ..... A [2]

2. at  $24\Omega$ .

current = ..... A [2]

**(ii)** Use your answers in **(b)(i)** to calculate the change in the total power produced by the battery when the resistance of  $R_3$  is changed from zero to  $24\Omega$ .

change in power = ..... W [2]

**(c)** Switch S in Fig. 6.1 is now **opened**.

Resistors  $R_1$  and  $R_2$  are made from metal wires. Some data for these resistors are shown in Fig. 6.2.

|                                                   | $R_1$ | $R_2$   |
|---------------------------------------------------|-------|---------|
| cross-sectional area of wire                      | $A$   | $1.8A$  |
| number of free electrons per unit volume in metal | $n$   | $0.50n$ |

**Fig. 6.2**

Determine the ratio

$$\frac{\text{average drift speed of free electrons in } R_1}{\text{average drift speed of free electrons in } R_2}$$

ratio = ..... [2]

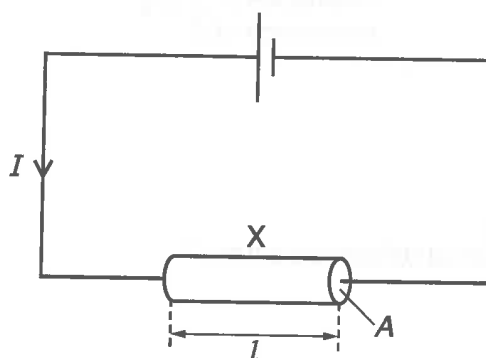
**47. M/J 16/P21/Q6**

booklet (scan + OCR)

**(a)** Define the *coulomb*.

.....[1]

**(b)** A resistor X is connected to a cell as shown in Fig. 6.1.



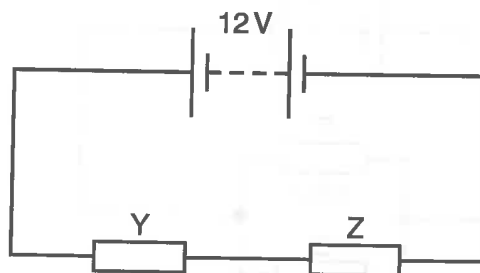
**Fig. 6.1**

The resistor is a wire of cross-sectional area  $A$  and length  $l$ . The current in the wire is  $I$ . Show that the average drift speed  $v$  of the charge carriers in X is given by the equation

$$v = \frac{I}{nAe}$$

where  $e$  is the charge on a charge carrier and  $n$  is the number of charge carriers per unit volume in X.

**(c)** A 12V battery with negligible internal resistance is connected to two resistors Y and Z, as shown in Fig. 6.2. [3]



**Fig. 6.2**

The resistors are made from wires of the same material. The wire of Y has a diameter  $d$  and length  $l$ . The wire of Z has a diameter  $2d$  and length  $2l$ .

(i) Determine the ratio

$$\frac{\text{average drift speed of the charge carriers in Y}}{\text{average drift speed of the charge carriers in Z}} .$$

$$\text{ratio} = \dots\dots\dots [3]$$

(ii) Show that

$$\frac{\text{resistance of Y}}{\text{resistance of Z}} = 2.$$

[2]

(iii) Determine the potential difference across Y.

$$\text{potential difference} = \dots\dots\dots \text{ V } [2]$$

(iv) Determine the ratio

$$\frac{\text{power dissipated in Y}}{\text{power dissipated in Z}} .$$

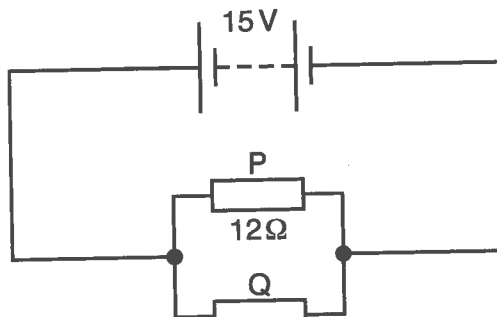
$$\text{ratio} = \dots\dots\dots [1]$$

**48. M/J 16/P23/Q6**

booklet (scan + OCR)

**(a)** Define the *ohm*.

..... [1]

**(b)** A 15V battery with negligible internal resistance is connected to two resistors P and Q, as shown in Fig. 6.1.**Fig. 6.1**

The resistors are made of wires of the same material. The wire of P has diameter  $d$  and length  $2l$ . The wire of Q has diameter  $2d$  and length  $l$ .  
The resistance of P is  $12\Omega$ .

**(i)** Show that the resistance of Q is  $1.5\Omega$ .**(ii)** Calculate the total power dissipated in the resistors P and Q. [3]

power = .....W [3]

**(iii)** Determine the ratio

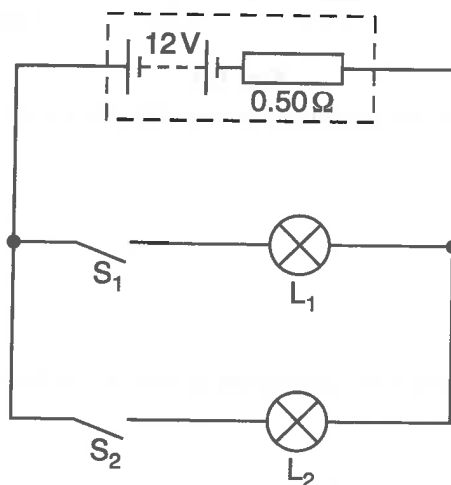
$$\frac{\text{average drift speed of the charge carriers in P}}{\text{average drift speed of the charge carriers in Q}}$$

ratio = ..... [3]

**49. O/N 15/P22/Q6**

booklet (scan + OCR)

A 12V battery with internal resistance  $0.50\Omega$  is connected to two identical filament lamps  $L_1$  and  $L_2$  as shown in Fig. 6.1.

**Fig. 6.1**

The lamps are connected to the battery via switches  $S_1$  and  $S_2$ . The power rating of each lamp is 48W for a potential difference of 12V.

(a)  $S_1$  is closed and  $S_2$  open.

State and explain whether the power transformed in  $L_1$  is 48W.

.....  
 ..... [2]

(b)  $S_2$  is now also closed.

(i) State and explain the effect on the current in  $L_1$ .

..... [1]

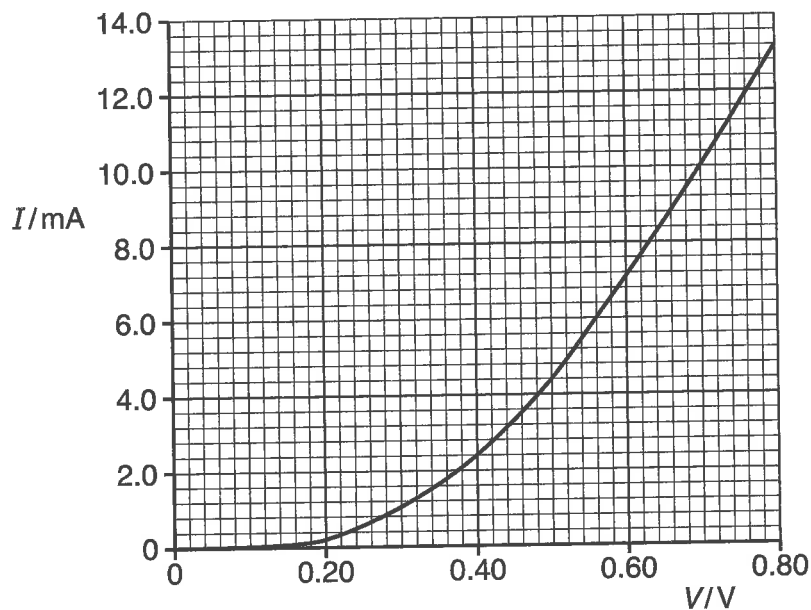
(ii) State and explain the effect on the resistance of  $L_1$ .

..... [1]

**50. O/N 15/P23/Q5**

booklet (scan + OCR)

(a) The  $I$ - $V$  characteristic of a semiconductor diode is shown in Fig. 5.1.

**Fig. 5.1**

- (i) Use Fig. 5.1 to explain the variation of the resistance of the diode as  $V$  increases from zero to 0.8 V.

.....  
.....  
..... [3]

- (ii) Use Fig. 5.1 to determine the resistance of the diode for a current of 4.4 mA.

resistance = .....  $\Omega$  [2]

- (b) A cell of e.m.f. 1.2V and negligible internal resistance is connected in series to a semiconductor diode and a resistor  $R_1$ , as shown in Fig. 5.2.

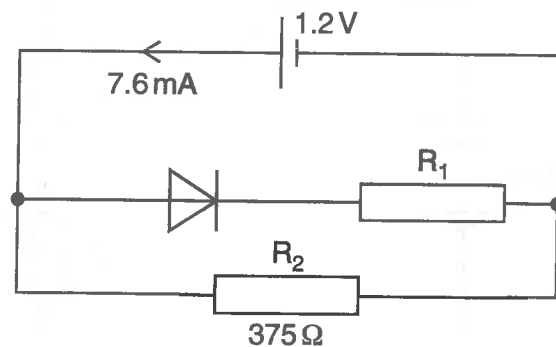


Fig. 5.2

A resistor  $R_2$  of resistance  $375\Omega$  is connected across the cell.

The diode has the characteristic shown in Fig. 5.1. The current supplied by the cell is 7.6mA. Calculate

- (i) the current in  $R_2$ ,

current = ..... A [1]

- (ii) the resistance of  $R_1$ ,

resistance = .....  $\Omega$  [2]

- (iii) the ratio

$$\frac{\text{power dissipated in the diode}}{\text{power dissipated in } R_2}.$$

ratio = ..... [2]

### 51. M/J 15/P22/Q5

booklet (scan + OCR)

- (a) On Fig. 5.1, sketch the temperature characteristic of a thermistor.

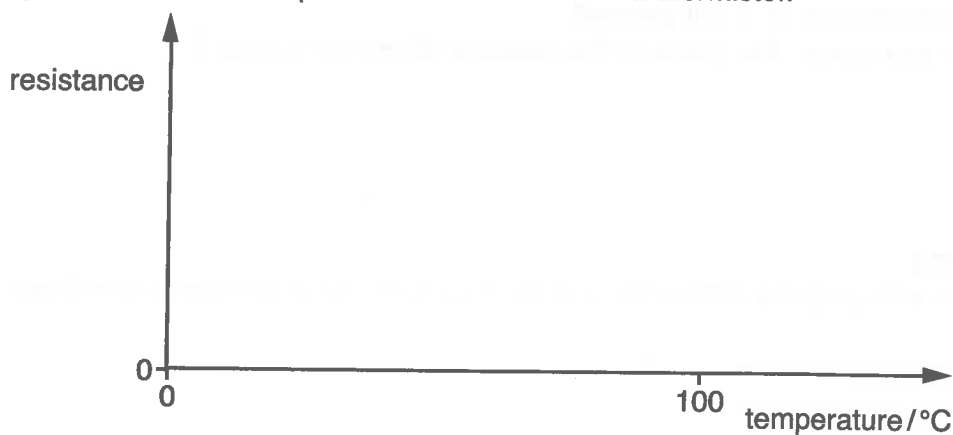
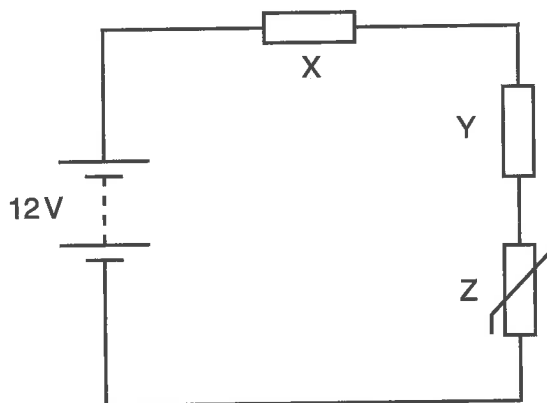


Fig. 5.1

[2]

(b) A potential divider circuit is shown in Fig. 5.2.



**Fig. 5.2**

The battery of electromotive force (e.m.f.) 12 V and negligible internal resistance is connected in series with resistors X and Y and thermistor Z. The resistance of Y is  $15\text{ k}\Omega$  and the resistance of Z at a particular temperature is  $3.0\text{ k}\Omega$ . The potential difference (p.d.) across Y is 8.0 V.

(i) Explain why the power transformed in the battery equals the total power transformed in X, Y and Z.

..... [1]

(ii) Calculate the current in the circuit.

current = ..... A [2]

(iii) Calculate the resistance of X.

resistance = .....  $\Omega$  [3]

(iv) The temperature of Z is increased.

State and explain the effect on the potential difference across Z.

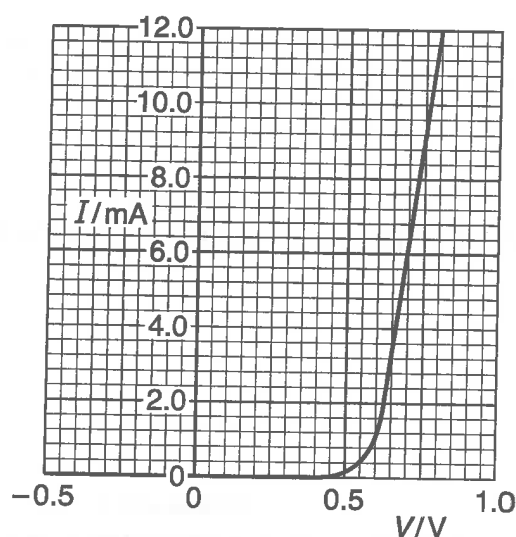
.....

..... [2]

## 52. M/J 15/P21/Q5

booklet (scan + OCR)

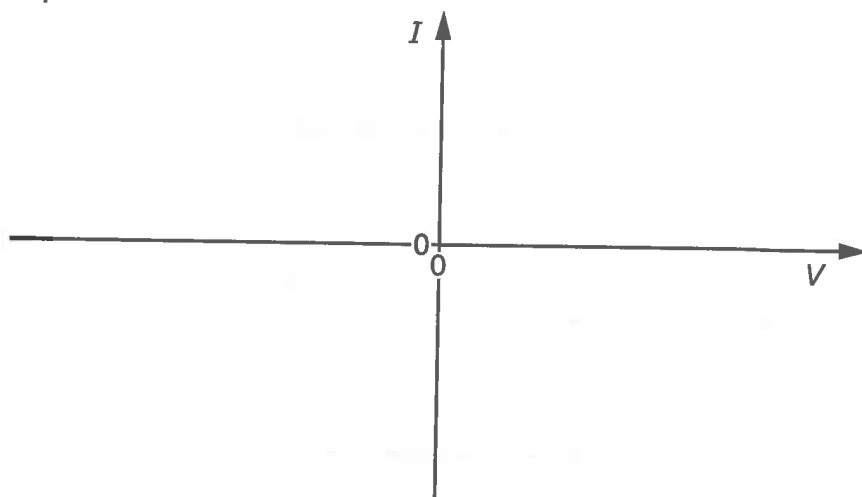
The variation with potential difference (p.d.)  $V$  of current  $I$  for a semiconductor diode is shown in Fig. 5.1.

**Fig. 5.1**

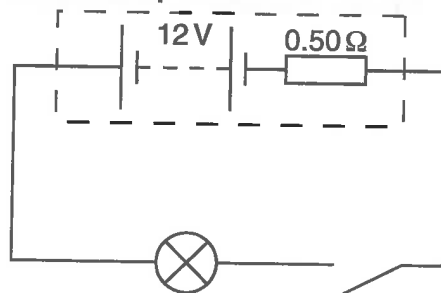
- (a) Use Fig. 5.1 to describe the variation of the resistance of the diode between  $V = -0.5\text{ V}$  and  $V = 0.8\text{ V}$ .

.....  
 ..... [2]

- (b) On Fig. 5.2, sketch the variation with p.d.  $V$  of current  $I$  for a filament lamp. Numerical values are not required.

**Fig. 5.2**

- (c) Fig. 5.3 shows a power supply of electromotive force (e.m.f.)  $12\text{ V}$  and internal resistance  $0.50\ \Omega$  connected to a filament lamp and switch.

**Fig. 5.3**

The filament lamp has a power of 36W when the p.d. across it is 12V.  
(i) Calculate the resistance of the lamp when the p.d. across it is 12V.

resistance = .....  $\Omega$  [1]

(ii) The switch is closed and the current in the lamp is 2.8 A. Calculate the resistance of the lamp.

resistance = .....  $\Omega$  [3]

(d) Explain how the two values of resistance calculated in (c) provide evidence for the shape of the sketch you have drawn in (b).

..... [1]

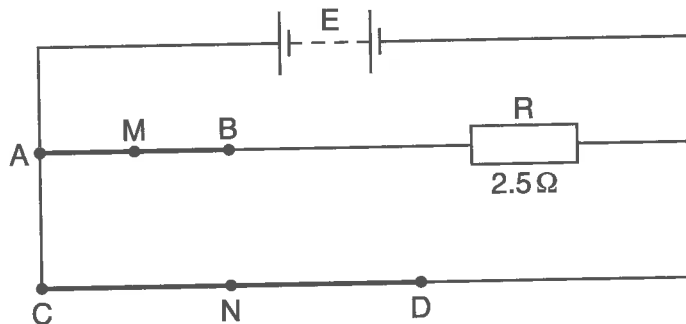
**53. M/J 15/P23/Q5**

booklet (scan + OCR)

A uniform resistance wire AB has length 50cm and diameter 0.36mm. The resistivity of the metal of the wire is  $5.1 \times 10^{-7} \Omega \text{ m}$ .

(a) Show that the resistance of the wire AB is  $2.5 \Omega$ .

(b) The wire AB is connected in series with a power supply E and a resistor R as shown in Fig. 5.1. [2]

**Fig. 5.1**

The electromotive force (e.m.f.) of E is 6.0V and its internal resistance is negligible. The resistance of R is  $2.5 \Omega$ . A second uniform wire CD is connected across the terminals of E. The wire CD has length 100cm, diameter 0.18mm and is made of the same metal as wire AB. Calculate

(i) the current supplied by E,

current = ..... A [4]

(ii) the power transformed in wire AB,

power = ..... W [2]

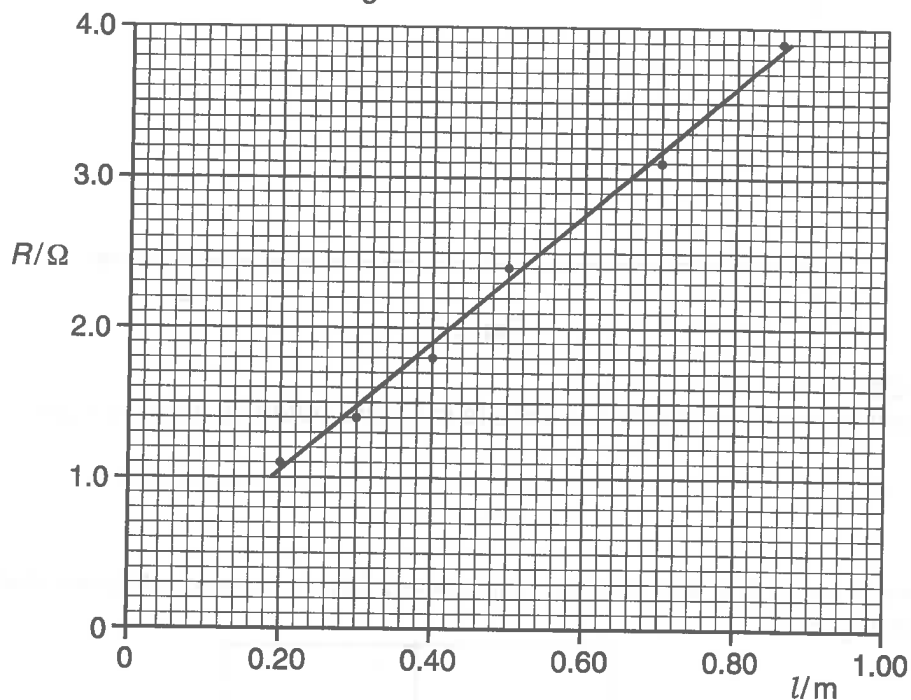
(iii) the potential difference (p.d.) between the midpoint M of wire AB and the midpoint N of wire CD.

p.d. = ..... V [2]

**54. O/N 14/P21/Q3**

booklet (scan + OCR)

The resistance  $R$  of a uniform metal wire is measured for different lengths  $l$  of the wire. The variation with  $l$  of  $R$  is shown in Fig. 3.1.

**Fig. 3.1**

- (a) The points shown in Fig. 3.1 do not lie on the best-fit line. Suggest a reason for this.

..... [1]

- (b) Determine the gradient of the line shown in Fig. 3.1.

gradient = ..... [2]

- (c) The cross-sectional area of the wire is  $0.12 \text{ mm}^2$ .

Use your answer in **(b)** to determine the resistivity of the metal of the wire.

resistivity = .....  $\Omega \text{ m}$  [3]

- (d)** The resistance  $R$  of different wires is measured. The wires are of the same metal and same length but have different cross-sectional areas  $A$ .

On Fig. 3.2, sketch a graph to show the variation with  $A$  of  $R$ .

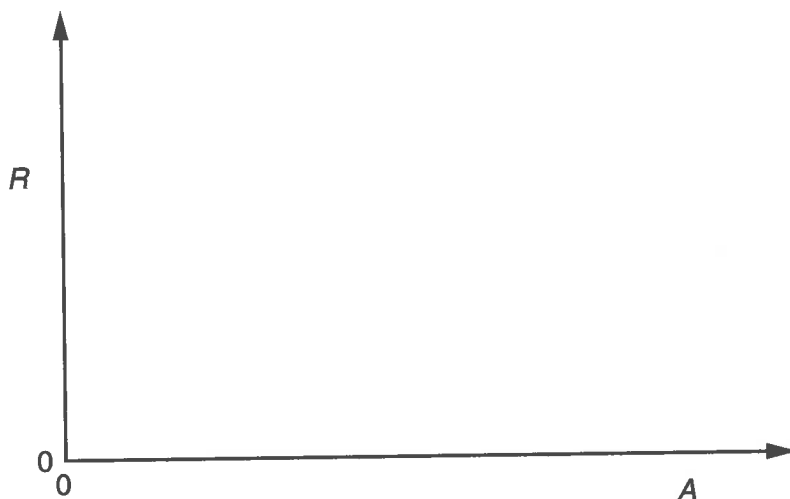


Fig. 3.2

[2]

### 55. M/J 14/P21/Q6

booklet (scan + OCR)

- (a)** Distinguish between *electromotive force* (e.m.f.) and *potential difference* (p.d.).

.....  
 ..... [2]

- (b)** A battery of e.m.f. 12V and internal resistance  $0.50\Omega$  is connected to two identical lamps, as shown in Fig. 6.1.

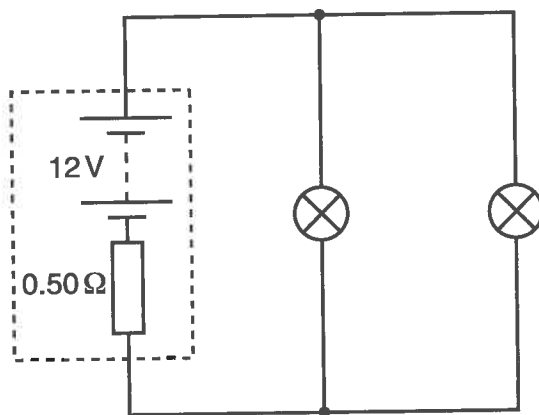


Fig. 6.1

Each lamp has constant resistance. The power rating of each lamp is 48W when connected across a p.d. of 12V.

- (i) Explain why the power dissipated in each lamp is not 48W when connected as shown in Fig. 6.1.

..... [1]

- (ii) Calculate the resistance of one lamp.

resistance = .....  $\Omega$  [2]

- (iii) Calculate the current in the battery.

current = ..... A [2]

- (iv) Calculate the power dissipated in one lamp.

power = ..... W [2]

- (c) A third identical lamp is placed in parallel with the battery in the circuit of Fig. 6.1. Describe and explain the effect on the terminal p.d. of the battery.

.....  
..... [2]

## Answers Section

### 1. M/J 26/P22/Q5/b(ii)

|          |                                                                                                            |           |                                                                                                                                                                                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------------|-----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5(b)(ii) | $v = I / Anq$<br>$= (8.6 / 2) / (6.1 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$ | <b>C1</b> | no mark for symbol formula as given in the formula sheet<br><br>ECF current from <b>5(b)(i)</b><br>Allow $(12 / 1.4) / (2 \times 6.1 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$ if treats as a single wire of double area                 |
|          | $= 5.2 \times 10^{-5} \text{ m s}^{-1}$                                                                    | <b>A1</b> | <u>Special case:</u><br>candidates that do not half their current from <b>5(b)(i)</b> can be credited 1/2 if working is shown e.g.<br>$= 8.6 / (6.1 \times 10^{-6} \times 8.5 \times 10^{28} \times 1.6 \times 10^{-19})$<br>$= 1.0 \times 10^{-4} \text{ m s}^{-1}$ |

### 2. M/J 26/P23/Q4/b(ii,iv)

|          |                                                                                                                                                                                |           |           |           |    |                                                 |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-----------|-----------|----|-------------------------------------------------|
| 4(b)(ii) | $P = V^2 / R$ or $P = VI$ <u>and</u> $V = IR$ or $P = I^2R$ <u>and</u> $V = IR$<br><br>= $4.78^2 / 230$<br>or = $4.78 \times (4.78 / 230)$<br>or = $(4.78 / 230)^2 \times 230$ |           |           | C1        |    |                                                 |
|          | = 0.099 W                                                                                                                                                                      |           |           | A1        |    |                                                 |
| 4(b)(iv) |                                                                                                                                                                                | decreases | unchanged | increases | B1 | Allow symbols other than ticks if no ticks seen |
|          | resistance of LDR                                                                                                                                                              |           |           | ✓         |    |                                                 |
|          | current in cell                                                                                                                                                                | ✓         |           |           | B1 |                                                 |
|          | p.d. across LDR                                                                                                                                                                |           |           | ✓         | B1 |                                                 |

### 3. M/J 26/P24/Q5/a,b,d

|      |                                  |           |                                                                                                                                                                                    |
|------|----------------------------------|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5(a) | (semiconductor) diode            | <b>A1</b> | Allow any other type of diode. E.g. LED                                                                                                                                            |
| 5(b) | straight line through the origin | <b>B1</b> | Tolerances $\pm \frac{1}{2}$ small square for both marks                                                                                                                           |
|      | a line passing through (0.8, 20) | <b>B1</b> | Ignore line continuing outside graph past 0.80 V<br><br>Check for a candidate's graph drawn in space below Fig. 5.1. Allow one mark for straight line from a labelled origin (0,0) |
| 5(d) | 0 A                              | <b>A1</b> | Allow 'approximately zero' but not '-'                                                                                                                                             |

### 4. M/J 26/P24/Q6/a,b,c

|      |                                                                               |             |                                                                                                     |
|------|-------------------------------------------------------------------------------|-------------|-----------------------------------------------------------------------------------------------------|
| 6(a) | $P = V^2 / R$                                                                 | <b>C1</b>   | Allow any subject.<br>Must see full working and answer in all cases.<br>Not 1 kW need $1000 / 10^3$ |
|      | so $R = (V^2 / P) = 230^2 / 1000 = 53 \Omega$                                 | <b>A1</b>   |                                                                                                     |
|      | Or<br>$P = IV$ <u>and</u> $V = IR$                                            | <b>(C1)</b> |                                                                                                     |
|      | $R = 230 / (1000 / 230) = 53 \Omega$                                          | <b>(A1)</b> |                                                                                                     |
|      | Or<br>$P = IV$ <u>and</u> $P = I^2 R$                                         | <b>(C1)</b> |                                                                                                     |
|      | $R = 1000 / (1000 / 230)^2 = 53 \Omega$                                       | <b>(A1)</b> |                                                                                                     |
| 6(b) | $R = \rho L / A$<br>$L = (53 \times 5.7 \times 10^{-8}) / 1.3 \times 10^{-6}$ | <b>C1</b>   | Allow any subject.                                                                                  |
|      | $= 2.3 \text{ m}$                                                             | <b>A1</b>   |                                                                                                     |

|      |                                                                                        |           |                                          |
|------|----------------------------------------------------------------------------------------|-----------|------------------------------------------|
| 6(c) | $n = I / qAv$<br>$I = 1000 / 230 \text{ or } 230 / 53$<br>$= 4.35$                     | <b>C1</b> | Allow current of 4.3 to imply the C mark |
|      | $n = 4.35 / (1.6 \times 10^{-19} \times 5.7 \times 10^{-8} \times 5.4 \times 10^{-3})$ | <b>C1</b> |                                          |
|      | $= 8.8 \times 10^{28} \text{ m}^{-3}$                                                  | <b>A1</b> |                                          |

**5. MAR 26/P22/Q6/a,c(i,iii)**

|           |                                                                                                                                                         |           |
|-----------|---------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 6(a)      | a flow of charge carriers                                                                                                                               | <b>B1</b> |
| 6(c)(i)   | $\rho = RA / L$<br>$= 0.82 \times 2.0 \times 10^{-7} / 6.8$                                                                                             | <b>C1</b> |
|           | $= 2.4 \times 10^{-8} \Omega \text{ m}$                                                                                                                 | <b>A1</b> |
| 6(c)(iii) | $P = IV \text{ or } P = V^2 / R \text{ or } P = I^2 R$<br>$= 1.8 \times (1.50 - 0.15) \text{ or } (1.50 - 0.15)^2 / 0.75 \text{ or } 1.8^2 \times 0.75$ | <b>C1</b> |
|           | $= 2.4 \text{ W}$                                                                                                                                       | <b>A1</b> |

**6. O/N 25/P21/Q3/b(i),c(i),d(i)**

|         |                                                                                                |           |
|---------|------------------------------------------------------------------------------------------------|-----------|
| 3(b)(i) | $R_0 = \rho L / A$                                                                             | <b>A1</b> |
| 3(c)(i) | straight line with positive gradient starting at (0, $R_0$ )                                   | <b>B1</b> |
| 3(d)(i) | $L = [0.034 \times \pi \times (0.80 \times 10^{-3})^2] / 1.8 \times 10^{-8} = 3.8 \text{ (m)}$ | <b>A1</b> |

**7. O/N 25/P21/Q5/b(i)**

|         |                                                                                                                |           |
|---------|----------------------------------------------------------------------------------------------------------------|-----------|
| 5(b)(i) | line with negative gradient                                                                                    | <b>C1</b> |
|         | line from 0 °C to 100 °C, starting at (0, $R_0$ ) with negative gradient throughout and never reaching $R = 0$ | <b>A1</b> |

**8. O/N 25/P22/Q5/b(iii)**

|           |                                                                                                                                                |           |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(b)(iii) | $P = IV \text{ or } P = I^2 R \text{ or } P = V^2 / R$                                                                                         | <b>C1</b> |
|           | $P = 0.016 \times 4.8$<br><b>or</b><br>$P = 0.016^2 \times (4.8 / 0.016)$<br><b>or</b><br>$P = 4.8^2 / (4.8 / 0.016)$<br>$P = 0.077 \text{ W}$ | <b>A1</b> |

**9. O/N 25/P24/Q5/a(i),b(i,ii)**

|          |                                                                             |           |
|----------|-----------------------------------------------------------------------------|-----------|
| 5(a)(i)  | potential difference per unit current                                       | <b>B1</b> |
| 5(b)(i)  | (resistance) = $1.38 / 0.276 = 5.00 \text{ (}\Omega\text{)}$                | <b>A1</b> |
| 5(b)(ii) | $\rho = RA / L$                                                             | <b>C1</b> |
|          | $\rho = 5.00 \times \pi \times (0.496 \times 10^{-3})^2 / (4 \times 0.864)$ | <b>C1</b> |
|          | $= 1.12 \times 10^{-6} \Omega \text{ m}$                                    | <b>A1</b> |

**10. M/J 25/P21/Q6/a,b(ii)**

|           |                                                                                                                                                  |                                     |
|-----------|--------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 6(a)(i)   | $I = 1.3 / 1.1$<br>$= 1.2 \text{ A}$                                                                                                             | <b>A1</b>                           |
| 6(a)(ii)  | $v = I / nqA$<br>$= 1.2 / (8.5 \times 10^{28} \times 1.60 \times 10^{-19} \times 4.7 \times 10^{-7})$<br>$= 1.9 \times 10^{-4} \text{ m s}^{-1}$ | <b>C1</b><br><b>A1</b>              |
| 6(a)(iii) | $\rho = RA / L$<br>$= (1.1 \times 4.7 \times 10^{-7}) / 0.45$<br>$= 1.1 \times 10^{-6} \Omega \text{ m}$                                         | <b>C1</b><br><b>C1</b><br><b>A1</b> |
| 6(b)(ii)  | (The average drift speed will be) the same because the current is the same (in X).                                                               | <b>B1</b>                           |

**11. M/J 25/P22/Q6/a,b(i)**

|         |                                                                 |                        |
|---------|-----------------------------------------------------------------|------------------------|
| 6(a)    | energy transferred (to the component) per (unit) charge         | <b>B1</b>              |
| 6(b)(i) | $Q = It$<br>$= 0.030 \times 4.0 \times 60$<br>$= 7.2 \text{ C}$ | <b>C1</b><br><b>A1</b> |

**12. M/J 25/P23/Q7/a**

|      |                                                                                                   |                                     |
|------|---------------------------------------------------------------------------------------------------|-------------------------------------|
| 7(a) | $R = \rho L / A$<br>$= (1.12 \times 10^{-6} \times 1.5) / 2.45 \times 10^{-7}$<br>$= 6.86 \Omega$ | <b>C1</b><br><b>C1</b><br><b>A1</b> |
|------|---------------------------------------------------------------------------------------------------|-------------------------------------|

**13. M/J 25/P24/Q5/a,b**

|          |                                                                                                                                                                                                                                                                                                                               |                                     |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| 5(a)(i)  | temperature decreases, resistance decreases                                                                                                                                                                                                                                                                                   | <b>B1</b>                           |
| 5(a)(ii) | line from origin with decreasing gradient drawn in first quadrant<br>or line from origin with decreasing gradient drawn in third quadrant<br><br>line drawn in the third or first quadrant of similar composition (straight line from origin followed by correct curve) and similar size as compared by eye to the first line | <b>M1</b><br><b>A1</b>              |
| 5(b)(i)  | $E = P / I$<br>$= 18 / 1.5$<br>$= 12 \text{ V}$                                                                                                                                                                                                                                                                               | <b>C1</b><br><b>A1</b>              |
| 5(b)(ii) | $I = 3.3 - 1.5$<br>$= 1.8$<br><br>$I = Anvq$<br>$1.8 = 1.4 \times 10^{-9} \times 3.4 \times 10^{28} \times v \times 1.6 \times 10^{-19}$<br><br>$v = 0.24 \text{ m s}^{-1}$                                                                                                                                                   | <b>C1</b><br><b>C1</b><br><b>A1</b> |

**14. MAR 25/P22/Q6/a,b,c**

|         |                                                                                                                         |                        |
|---------|-------------------------------------------------------------------------------------------------------------------------|------------------------|
| 6(a)    | $V = IR$<br>$= 0.85 \times 3.3 \times 10^{-3}$<br>$= 2.8 \times 10^{-3} \text{ V}$                                      | <b>C1</b><br><b>A1</b> |
| 6(b)(i) | $(A =) \rho L / R$<br>$= 1.8 \times 10^{-8} \times 0.24 / 3.3 \times 10^{-3} = 1.3 \times 10^{-6} \text{ (m}^2\text{)}$ | <b>C1</b><br><b>A1</b> |

|           |                                                                                                            |             |
|-----------|------------------------------------------------------------------------------------------------------------|-------------|
| 6(b)(ii)  | $(n =) 2.6 \times 10^{22} / (1.3 \times 10^{-6} \times 0.24) = 8.3 \times 10^{28} \text{ (m}^{-3}\text{)}$ | <b>A1</b>   |
| 6(b)(iii) | $v = I / nAq$                                                                                              | <b>C1</b>   |
|           | $= 0.85 / (8.3 \times 10^{28} \times 1.3 \times 10^{-6} \times 1.6 \times 10^{-19})$                       |             |
|           | $= 4.9 \times 10^{-5} \text{ m s}^{-1}$                                                                    | <b>A1</b>   |
|           | OR                                                                                                         | <b>(C1)</b> |
|           | $v = IL / Nq$                                                                                              |             |
|           | $= 0.85 \times 0.24 / (2.6 \times 10^{22} \times 1.6 \times 10^{-19})$                                     |             |
|           | $= 4.9 \times 10^{-5} \text{ m s}^{-1}$                                                                    | <b>(A1)</b> |
| 6(c)(i)   | Length (of Q) is greater (than P)                                                                          | <b>B1</b>   |
|           | (Average cross-sectional) area (of Q) is less (than P)                                                     | <b>B1</b>   |
|           | Resistance is proportional to length / (cross-sectional) area                                              | <b>M1</b>   |
|           | (so) the resistance (of Q) is greater (than P)                                                             | <b>A1</b>   |
| 6(c)(ii)  | A line starting from a non-zero value of drift speed at distance = 0                                       | <b>B1</b>   |
|           | A line with an increasing positive gradient                                                                | <b>B1</b>   |

**15. O/N 24/P21/Q3/c(iv)**

|          |                                                                          |           |
|----------|--------------------------------------------------------------------------|-----------|
| 3(c)(iv) | $P = I^2 R$ and $R = \rho L / A$                                         | <b>C1</b> |
|          | $280 = I^2 \times (1.7 \times 10^{-8} \times 23) / (2.6 \times 10^{-8})$ | <b>C1</b> |
|          | $I = 4.3 \text{ A}$                                                      | <b>A1</b> |

**16. O/N 24/P22/Q5/a,b(ii,v,vi)**

|          |                                                                                                                         |           |
|----------|-------------------------------------------------------------------------------------------------------------------------|-----------|
| 5(a)     | energy transferred per unit charge (from electrical to other forms)                                                     | <b>B1</b> |
| 5(b)(ii) | $(V =) 7.0 \times 0.86 = 6.0 \text{ (V)}$                                                                               | <b>A1</b> |
| 5(b)(v)  | $P = IV$ or $P = I^2 R$ or $P = V^2 / R$                                                                                | <b>C1</b> |
|          | $P = 5.7 \times 210$<br>or<br>$P = 5.7^2 \times (210 / 5.7)$<br>or<br>$P = 210^2 / (210 / 5.7)$<br>$P = 1200 \text{ W}$ | <b>A1</b> |
| 5(b)(vi) | % efficiency = <u>useful</u> power out / (total) power in ( $\times 100$ )                                              | <b>C1</b> |
|          | $= 1200 / (230 \times 7.0) (\times 100)$                                                                                | <b>A1</b> |
|          | $= 0.75 (\times 100)$                                                                                                   |           |
|          | $= 75\% \text{ or } 74\% \text{ (using 3 s.f. value from (v) gives 74\%)}$                                              |           |

**17. O/N 24/P23/Q6/a,b**

|         |                                            |           |
|---------|--------------------------------------------|-----------|
| 6(a)    | potential difference per unit current      | <b>B1</b> |
| 6(b)(i) | $\rho = RA / L$                            | <b>C1</b> |
|         | $= (0.33 \times 8.0 \times 10^{-6}) / 2.4$ | <b>A1</b> |
|         | $= 1.1 \times 10^{-6} \Omega \text{ m}$    |           |

|           |                                                                                         |           |
|-----------|-----------------------------------------------------------------------------------------|-----------|
| 6(b)(ii)  | $Q = It$                                                                                | <b>C1</b> |
|           | $= 4.7 \times 5.0 \times 60$                                                            | <b>A1</b> |
|           | $= 1400 \text{ C}$                                                                      |           |
| 6(b)(iii) | $I = nAvq$                                                                              | <b>C1</b> |
|           | $n = 4.7 / (8.0 \times 10^{-6} \times 0.16 \times 10^{-3} \times 1.60 \times 10^{-19})$ |           |
|           | $= 2.3 \times 10^{28} \text{ m}^{-3}$                                                   | <b>A1</b> |

**18. M/J 24/P22/Q3/a,b,c**

|          |                                                                                                                                                                                                                                                                               |           |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 3(a)     | $Q = It$                                                                                                                                                                                                                                                                      | <b>C1</b> |
|          | $= 3.3 \times 10^4 \times 2.6 \times 10^{-5}$                                                                                                                                                                                                                                 | <b>A1</b> |
|          | $= 0.86 \text{ C}$                                                                                                                                                                                                                                                            |           |
| 3(b)     | $P = IV$<br>or<br>$P = VQ / t$<br>or<br>$V = IR$ and $P = V^2 / R$ or $P = I^2 R$                                                                                                                                                                                             | <b>C1</b> |
|          | $P = 3.3 \times 10^4 \times 3.0 \times 10^7$<br>or<br>$P = (3.0 \times 10^7 \times 0.86) / (2.6 \times 10^{-5})$<br>or<br>$P = (3.0 \times 10^7)^2 / 910$<br>or<br>$P = (3.3 \times 10^4)^2 \times 910$<br><br>$P = 9.9 \times 10^{11} \text{ (W)}$<br><br>$= 990 \text{ GW}$ | <b>A1</b> |
|          |                                                                                                                                                                                                                                                                               |           |
| 3(c)(i)  | $R = \rho L / A$                                                                                                                                                                                                                                                              | <b>C1</b> |
|          | $9.6 = 1.7 \times 10^{-8} \times 95 / \pi r^2$                                                                                                                                                                                                                                | <b>C1</b> |
|          | $r = 2.3 \times 10^{-4} \text{ m}$                                                                                                                                                                                                                                            | <b>A1</b> |
| 3(c)(ii) | (resistance) decreases by a factor of four                                                                                                                                                                                                                                    | <b>A1</b> |

**19. MAR 24/P22/Q7/a,c(ii)**

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |           |
|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 7(a)     | energy (transferred) per (unit) charge                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | <b>B1</b> |
| 7(c)(ii) | Any 3 from: <ul style="list-style-type: none"> <li>before <math>t_1</math> / when current constant, the (total) resistance is constant</li> <li>at <math>t_1</math> / when current increases, the (total) resistance decreases (due to decrease of external resistance)</li> <li>(after <math>t_1</math>) temperature (of lamp) increases (so the resistance of the lamp increases)</li> <li>(after <math>t_1</math>) resistance <b>of lamp</b> increases (so total resistance increases so the current in the ammeter decreases)</li> </ul> | <b>B3</b> |

**20. O/N 23/P21/Q6/a(i,iii)**

|         |                                      |           |
|---------|--------------------------------------|-----------|
| 6(a)(i) | $(I =) 16 / 320 = 0.050 \text{ (A)}$ | <b>A1</b> |
|---------|--------------------------------------|-----------|

|           |                                                                                                                                                                                                                                    |           |
|-----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 6(a)(iii) | $P = V^2 / R$ or $P = VI$ or $P = I^2 R$                                                                                                                                                                                           | <b>C1</b> |
|           | $\text{ratio} = (9^2 / 720) / (9^2 / 240)$<br>or<br>$\text{ratio} = (9 \times 0.0125) / (9 \times 0.0375)$<br>or<br>$\text{ratio} = (0.0125^2 \times 720) / (0.0375^2 \times 240)$<br>$\text{ratio} = 0.1125 / 0.3375$<br>$= 0.33$ | <b>A1</b> |

**21. O/N 23/P22/Q6/a,b(i,ii,iv),c(i)**

|          |                                                                                    |           |
|----------|------------------------------------------------------------------------------------|-----------|
| 6(a)     | $R = \rho L / A$                                                                   | <b>C1</b> |
|          | $\rho = 0.92 \times 5.3 \times 10^{-7}$<br>$= 4.9 \times 10^{-7} \Omega \text{ m}$ | <b>A1</b> |
| 6(b)(i)  | (current $\Rightarrow$ ) $6.4 / 1600 = 4.0 \times 10^{-3} \text{ (A)}$             | <b>A1</b> |
| 6(b)(ii) | charge $= 4.0 \times 10^{-3} \times 3.2 \times 60$<br>$(= 0.768 \text{ C})$        | <b>C1</b> |
|          | number $= 0.768 / 1.6 \times 10^{-19}$<br>$= 4.8 \times 10^{18}$                   | <b>A1</b> |
| 6(b)(iv) | $P = I^2 R$ or $P = VI$ or $P = V^2 / R$                                           | <b>C1</b> |
|          | $\text{ratio} = (6.4^2 / 1600) / [(4.0 \times 10^{-3})^2 \times 1400]$<br>$= 1.1$  | <b>A1</b> |
| 6(c)(i)  | increase                                                                           | <b>B1</b> |

**22. O/N 23/P23/Q7/a,b(i),c**

|          |                                                                                                                                                                                                                                   |           |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|
| 7(a)     | current (through a conductor is directly) proportional to potential difference (across the conductor) or vice versa                                                                                                               | <b>M1</b> |
|          | (provided that) temperature (of conductor remains) constant                                                                                                                                                                       | <b>A1</b> |
| 7(b)(i)  | $R = \rho L / A$                                                                                                                                                                                                                  | <b>C1</b> |
|          | $\rho = (18 \times 7.2 \times 10^{-8}) / 0.94$<br>$= 1.4 \times 10^{-6} \Omega \text{ m}$                                                                                                                                         | <b>A1</b> |
| 7(c)(i)  | $I = Anvq$                                                                                                                                                                                                                        | <b>C1</b> |
|          | $q = 0.93 / [(7.2 \times 10^{-8}) \times (9.0 \times 10^{28}) \times (1.3 \times 10^{-3})]$<br>$q = 1.1 \times 10^{-19} \text{ C}$                                                                                                | <b>A1</b> |
| 7(c)(ii) | charge / $q$ (value) is below $1.6 \times 10^{-19} \text{ (C)}$<br>or<br>charge cannot be below $1.6 \times 10^{-19} \text{ (C)}$<br>or<br>(the charge carriers / $q$ ) should have a charge of $1.6 \times 10^{-19} \text{ (C)}$ | <b>B1</b> |

**23. M/J 23/P22/Q6**

(a) temperature decreases (so) resistance decreases [1]

(b) (i) current =  $V/R$  [1]

(ii)  $I = nAvq$

$$n = N/V \text{ or } n = N/AL$$

$$v = (V/R) / [(V/L) (N/V) e] \text{ or } (V/R) / [A (N/AL) e]$$

$$= VL/RNe$$

or

$$v = L/t$$

$$= L/(Q/I)$$

$$= LI/Q$$

$$= L(V/R)/Ne$$

$$= VL/RNe$$

[2]

(iii) time = distance/speed or  $Q/I$

$$= L/(VL/RNe) \text{ or } Ne/(V/R)$$

$$\text{time} = RNe/V$$

[1]

**24. M/J 23/P21/Q6**

(a) (number density =)  $2.3 \times 10^{23} / (1.5 \times 10^{-6} \times 1.8) = 8.5 \times 10^{28} \text{ (m}^{-3}\text{)}$  [1]

(b)  $I = Q/t$  or  $I = 172/2.5 \times 60$  or  $I = 1.1(5)$

$$I = nAvq$$

$$172/(2.5 \times 60) = 1.5 \times 10^{-6} \times 8.5 \times 10^{28} \times v \times 1.6 \times 10^{-19}$$

$$v = 5.6 \times 10^{-5} \text{ m s}^{-1}$$

[3]

**25. M/J 23/P23/Q5**

(a) correct symbol for the heater or for the LDR

all correct symbols in series (ignore voltmeter) and no extra symbols

correct symbol for voltmeter and in parallel with the heater

[3]

(b) (i)  $R = \rho L/A$

$$= (1.1 \times 10^{-6} \times 2.0) / 3.8 \times 10^{-7}$$

$$= 5.8 \Omega$$

[2]

(ii)  $I = 4.8/5.8$

$$= 0.83 \text{ A}$$

[1]

(c) (i) A larger (for new wire)

or

$$A \propto d^2 \text{ (and } d \text{ larger for new wire)}$$

or

$$R \propto 1/d^2 \text{ (and } d \text{ larger for new wire)}$$

so  $R$  is less (than that of first wire)

[2]

(ii) (heater / total resistance decreases so) current (in circuit) increases

(so p.d. across LDR increases)

or

heater resistance decreases so it has a smaller share/proportion/fraction of the (total) voltage / e.m.f.

(so voltmeter) reading is less (than 4.8 V)

[2]

**26. O/N 22/P21/Q5**

- (a) current (through a conductor is directly) proportional to potential difference (across the conductor)  
(provided that) temperature (of conductor remains) constant [2]
- (b) (i) (ratio of)  $V/I$  increases (as p.d. increases) [1]  
(ii) (as p.d. increases, current increases so) temperature increases [1]
- (c) (i)  $I = 1.55 \text{ A}$  [1]  
(ii)  $P = VI$  or  $P = I^2 R$  or  $P = V^2 / R$   
 $= 6.0 \times 1.55 \times 2$  or  $1.55^2 \times 3.87 \times 2$  or  $(6.0^2 / 3.87) \times 2$   
 $= 19 \text{ W}$  [2]  
(iii)  $I = 1.78 - 1.55$   
 $(= 0.23 \text{ A})$   
 $R = 12.0 / 0.23$   
 $= 52 \Omega$  [2]
- (d) lamp P: p.d. across lamp decreases to zero so goes 'out'  
lamp Q: p.d. across lamp increases to 12 V so gets brighter [2]

**27. M/J 22/P23/Q6**

- (a) (i) line passes through (0,0) and is in first and third quadrants only  
gradient of line becoming less steep in both quadrants and roughly symmetrical [2]  
(ii) (as  $I$  increases) the temperature (of the filament wire/lamp) increases  
(as  $I$  / temperature /  $V$  increases) the resistance (of wire/lamp) increases  
(as  $I$  / temperature /  $V$  increases the graph curves because) ratio  $V/I$  increases or ratio  $I/V$  decreases [3]
- (b)  $R = \rho L / A$   
 $= (5.6 \times 10^{-8} \times 5.8) / 3.4 \times 10^{-8}$   
 $= 9.6 \Omega$  [2]

(c)

| position of switches |        | ammeter readings |              |              |
|----------------------|--------|------------------|--------------|--------------|
| $S_1$                | $S_2$  | $X/\text{A}$     | $Y/\text{A}$ | $Z/\text{A}$ |
| open                 | open   | 0                | 0            | 0            |
| open                 | closed | 1.0              | 0            | 1.0          |
| closed               | open   | 2.0              | 2.0          | 0            |
| closed               | closed | 3.0              | 2.0          | 1.0          |

second row: both values correct

third row: all three values correct

fourth row:  $X = Y + Z$  (any values)

all three values correct [4]

**28. MAR 22/P22/Q6**

- (a) (i)  $P = VI$   
 $I = 36 / 8.0$   
 $= 4.5 \text{ A}$  [2]

$$\begin{aligned}
 \text{(ii) charge} &= 4.5 \times 50 \\
 &= 225 \\
 \text{number} &= 225 / 1.6 \times 10^{-19} \\
 &= 1.4 \times 10^{21}
 \end{aligned}
 \quad [2]$$

$$\begin{aligned}
 \text{(iii) } R &= V^2 / P \quad \text{or} \quad R = V / I \quad \text{or} \quad R = P / I^2 \\
 &= 8.0^2 / 36 \quad \text{or} \quad = 8.0 / 4.5 \quad \text{or} \quad = 36 / 4.5^2 \\
 &= 1.8 \, \Omega
 \end{aligned}
 \quad [2]$$

$$\begin{aligned}
 \text{(b) } R &= \rho L / A \\
 L &= (1.8 \times 0.25 \times 10^{-6}) / 1.4 \times 10^{-6} \\
 &= 0.32 \, \text{m}
 \end{aligned}
 \quad [2]$$

(c) (larger cross-sectional area, same length, same resistivity and so) less resistance (same p.d. and more current so) more power (dissipated) [2]

(d) current (in wire) is the same (same p.d. across wire so) power stays the same [2]

### 29. M/J 21/P22/Q5

(a) volt / ampere [1]

$$\begin{aligned}
 \text{(b) } R &= \rho L / A \\
 A &= (V / L) \\
 \text{(so) } R &= \rho L^2 / V \text{ (with } \rho \text{ and } V \text{ constant so } R \propto L^2)
 \end{aligned}
 \quad [2]$$

$$\text{(c) (i) } E = 2.4 \, \text{V} \quad [1]$$

$$\begin{aligned}
 \text{(ii) } P &= VI \text{ or } I^2 R \text{ or } V^2 / R \\
 &= 1.3 \times 5.0 \text{ or } 5.0^2 \times 0.26 \text{ or } 1.3^2 / 0.26 \\
 &= 6.5 \, \text{W}
 \end{aligned}
 \quad [3]$$

$$\text{(iii) } (-) \text{ internal resistance or } (-) r \quad [1]$$

### 30. M/J 21/P23/Q5

(a) energy per unit charge  
energy transferred by source driving charge around the complete circuit  
or energy transferred from other forms to electrical energy [2]

(b) there is a p.d. across the internal resistance/ $r$   
change in current/ $I$  results in a change in p.d. across the internal resistance  
 $V = E - \text{p.d. across internal resistance}$  or  
change in p.d. across  $r$  causes a change in  $V$  (as e.m.f. is constant) [3]

$$\text{(c) (i) } E = 7.4 \, \text{V} \quad [1]$$

$$\text{(ii) maximum current} = 0.92 \, \text{A} \quad [1]$$

$$\begin{aligned}
 \text{(iii) } r &= E / I_{\text{MAX}} \text{ or } (-) \text{gradient} \\
 \text{e.g. } r &= 7.4 / 0.92 \\
 &= 8.0 \, \Omega
 \end{aligned}
 \quad [2]$$

(d) straight line with negative gradient that is smaller in magnitude than the original line  
line which would have intercept on  $V$ -axis below the original line [2]

### 31. MAR 21/P22/Q6/a,b(i,ii,iii)

(a) sum of current(s) into junction = sum of current(s) out of junction  
or (algebraic) sum of current(s) at a junction is zero [1]

$$\begin{aligned}
 \text{(b) (i) } I &= 3.6 - 2.1 \\
 &= 1.5
 \end{aligned}$$

$$V = 4.4$$

$$R = 4.4 / 1.5$$

$$= 2.9 \, \Omega$$

$$(ii) \quad 12.0 = 4.4 + 3.6r \text{ or } 12.0 = 3.6(1.2 + r) \quad [3]$$

$$r = 2.1 \, \Omega$$

$$(iii) \quad t = (470 \times 10^3 - 240 \times 10^3) / (12 \times 3.6) \quad [2]$$

$$= 5300 \, s$$

[2]

**32. O/N 20/P21/Q7**

(a) volt / ampere

$$(b) \quad R = \rho L / A \quad [1]$$

$$A = 460 \times 10^{-9} \times 2.5 / 3.2$$

$$= 3.6 \times 10^{-7} \, m^2$$

(c) (i) energy is dissipated in the internal resistance/ $r$  [3]

$$(ii) \quad E = IR + Ir \text{ or } E = I(R + r) \quad [1]$$

$$(iii) \quad P = I^2 R \text{ or } P = I^2 r \quad [1]$$

$$I = E / 2r$$

$$(\text{so}) \quad P = E^2 / 4r$$

[2]

**33. M/J 20/P21/Q5**

$$(a) (i) \quad R = \rho L / A$$

$$A = (2.6 \times 10^{-8} \times 59) / 3.4 = 4.5 \times 10^{-7} \, m^2$$

$$(ii) \quad I = 1.8 / 3.4 \quad [2]$$

$$= 0.53 \, A$$

$$(iii) \quad I = Anvq \quad [1]$$

$$v = 0.53 / (4.5 \times 10^{-7} \times 6.1 \times 10^{28} \times 1.60 \times 10^{-19})$$

$$= 1.2 \times 10^{-4} \, m \, s^{-1}$$

[2]

(b) (i) (cross-sectional) area/ $A$  is less

( $I$ ,  $n$ ,  $e$  the same so) average drift speed is greater

[2]

(ii) (area is less so) more resistance/ $R$

( $I$  is the same, so) more power/ $P$  or

$$(P = I^2 \rho L / A \text{ so}) \quad P \propto 1 / A$$

( $A$  is less so) more  $P$

[2]

(c) (i)  $180 \, \Omega$  and  $90 \, \Omega$  resistors shown connected in parallel

[1]

(ii) resistors connected in parallel labelled as  $180 \, \Omega$  and  $90 \, \Omega$  and

the other resistor labelled as  $30 \, \Omega$   $V_{OUT}$  or  $8.0 \, V$  labelled across the two resistors in parallel

[2]

**34. M/J 20/P23/Q5**

(a) joule per coulomb

[1]

$$(b) (i) \quad 1 / R = 1 / R_1 + 1 / R_2$$

$$= 1 / 300 + 1 / 200$$

$$R = 75 \, \Omega$$

$$(ii) \quad R = 75 + 55$$

$$= 130 \, \Omega$$

[1]

$$(c) (i) \quad 1. \quad P = I^2 R \text{ or}$$

$$P = VI \text{ and } V = IR$$

[1]

$$I = (0.20 / 55)^{0.5}$$

$$= 0.060 \text{ A}$$

$$2. I = 0.060 / 4$$

$$= 0.015 \text{ A}$$

[3]

$$(ii) \text{ potential difference} = 130 \times 0.060$$

$$= 7.8 \text{ V} \quad \text{or}$$

$$\text{potential difference} = (300 \times 0.015) + (55 \times 0.060)$$

$$= 7.8 \text{ V (other valid methods are also possible)}$$

[1]

### 35. O/N 19/P21/Q6

(a) work done / charge or

energy (transferred from electrical to other forms) / charge

[1]

(b) for  $V < 0.25 \text{ V}$  resistance is infinite/very high (as current is zero)

for  $V > 0.25 \text{ V}$  resistance decreases (as  $V$  increases)

[2]

(c) (i)  $R = V / I$

$$= 0.75 / (15 \times 10^{-3})$$

$$= 50 \Omega$$

[3]

$$(ii) 1. V_Y = 15 \times 10^{-3} \times 60 (= 0.90 \text{ V})$$

$$V_X = 2.0 - 0.90 - 0.75 (= 0.35 \text{ V})$$

$$R_X = 0.35 / (15 \times 10^{-3})$$

$$= 23 \Omega \quad \text{or}$$

$$\text{total } R = 60 + 50 + R_X$$

$$60 + 50 + R_X = 2.0 / (15 \times 10^{-3})$$

$$R_X = 23 \Omega$$

[3]

$$2. P = VI \quad \text{or} \quad P = EI \quad \text{or} \quad P = I^2 R \quad \text{or} \quad P = V^2 / R$$

$$\text{ratio} = \frac{(15 \times 10^{-3})^2 \times 60}{2.0 \times 15 \times 10^{-3}} \quad \text{or} \quad \frac{0.90 \times 15 \times 10^{-3}}{2.0 \times 15 \times 10^{-3}} \quad \text{or} \quad \frac{(0.90^2 / 60)}{2.0 \times 15 \times 10^{-3}}$$

$$= 0.45$$

[2]

### 36. M/J 19/P21/Q6

(a) energy is dissipated in the internal resistance

[1]

(b)  $E = V + Ir$

[1]

(c) (i) (graph shows) maximum value of potential difference is 2.8 (V) or

(graph shows) when current/ $I$  (from battery) is zero,  $V$  is 2.8 (V) /  $E$

[1]

(ii)  $r = (-)\text{gradient}$  or  $r = (E - V) / I$  or substituted values from the graph for  $E$ ,  $V$  and  $I$

$$r = 1.4 \Omega$$

[2]

(d) (i)  $R = 2.1 / 0.50$

$$= 4.2 \Omega$$

[1]

$$(ii) \text{ number} = 0.50 / 1.60 \times 10^{-19}$$

$$= 3.1 \times 10^{18}$$

[1]

(iii) energy =  $EIt$  or

$$P = EI \quad \text{and} \quad P = W / t$$

$$(9.2 - 1.6) \times 10^3 = 2.8 \times 0.50 \times t$$

$$t = 5.4 \times 10^3 \text{ s}$$

[3]

**37. MAR 19/P22/Q6****(a)** e.m.f.: energy transferred from chemical to electrical (per unit charge)

p.d.: energy transferred from electrical to thermal (per unit charge)

[2]

**(b) (i)** 1  $I = 4.8 / 32$ 

$$= 0.15 \text{ A}$$

[1]

$$2 \quad P = EI \quad \text{or} \quad P = VI \quad \text{or} \quad P = I^2 R \quad \text{or} \quad P = V^2 / R$$

$$= 6.0 \times 0.15 \quad \text{or} \quad 0.15^2 \times 40 \quad \text{or} \quad 6.0^2 / 40$$

$$= 0.90 \text{ W}$$

[2]

3 number =  $It / e$

$$= [0.15 \times 25] / 1.6 \times 10^{-19}$$

$$= 2.3 \times 10^{19} \quad \text{or}$$

$$Q = 0.15 \times 25 (= 3.75)$$

$$\text{number} = 3.75 / 1.6 \times 10^{-19}$$

$$= 2.3 \times 10^{19}$$

[2]

4  $4.8 / 6.0 = 32 / (R_{XY} + 32)$  or  $1.2 / 6.0 = R_{XY} / (R_{XY} + 32)$

or  $4.8 / 1.2 = 32 / R_{XY}$

$$R_{XY} = 8.0 \, \Omega$$

Alternative methods:

$$R_{XY} = (6.0 - 4.8) / 0.15 \text{ or}$$

$$= 8.0 \, \Omega \quad \text{or}$$

$$6.0 = 0.15 (32 + R_{XY})$$

$$R_{XY} = 40 - 32 = 8.0 \, \Omega$$

[2]

5  $1 / 8.0 = 1 / R_X + 1 / 24$

$$R_X = 12 \, \Omega$$

Alternative method:

$$I_Z = 4.8 / 32 = 0.15 \text{ and } I_Y = 1.2 / 24 = 0.05$$

$$I_X = 0.15 - 0.05 (= 0.10)$$

$$R_X = 1.2 / 0.10 = 12 \, \Omega$$

[2]

**(ii)** total resistance decreases

(so voltmeter) reading increases

[2]

**38. O/N 18/P22/Q7****(a)** A: (cross-sectional) area (of wire) $n$ : number of free electrons per unit volume or number density of free electrons

[2]

**(b)** line drawn between (X,  $v_x$ ) and (Y,  $4v_x$ )

line has increasing gradient

[2]

**39. O/N 18/P23/Q6****(a)** (coulomb is an) ampere second

[1]

**(b)**  $8.0 \times 10^{-19} \text{ C}$  and  $1.6 \times 10^{-19} \text{ C}$  both underlined (and no others underlined)

[1]

**(c)** line drawn between (S,  $1.00v_s$ ) and (T,  $0.25v_s$ )

line with decreasing magnitude of gradient

[2]

**40. M/J 18/P21/Q6****(a)** joule/coulomb

[1]

**(b) (i)** lamps have same p.d./lamps have p.d. of 2.7 V

$$\text{current} = 0.15 + 0.090 = 0.24 \text{ A}$$

[2]

**(ii)**  $R = (4.5 - 2.7) / 0.24$  or

$$R_P = 18 (\Omega) \text{ and } R_Q = 30 (\Omega)$$

$$I/R_T = 1/18 + 1/30 \text{ and so } R_T = 11.25$$

$$4.5 = 0.24 \times (R + 11.25)$$

$$R = 7.5 \Omega$$

[2]

(iii)  $R = \rho l / A$

$$R_P / R_Q = [(2.7/0.15)/(2.7/0.09)] (= 0.60)$$

$$\text{ratio} = 0.60 \times 2^2$$

$$= 2.4$$

[3]

(iv) less p.d. across resistor/greater p.d. across P  
greater current through P and so resistance (of P) increases

[2]

#### 41. O/N 17/P22/Q5

(a) (coulomb is) ampere second

[1]

(b) (i)  $E = V/d$  or  $E = F/Q$

$$F = VQ/d$$

$$F = (2.0 \times 10^2 \times 8.0 \times 10^{-19}) / 4.0 \times 10^{-2} = 4.0 \times 10^{-15} \text{ N}$$

[2]

(ii) arrow pointing to the left labelled 'electric force' and arrow pointing downwards labelled 'weight'

[1]

(iii) 1. resultant force  $= \sqrt{[(3.9 \times 10^{-15})^2 + (4.0 \times 10^{-15})^2]}$   
 $= 5.6 \times 10^{-15} \text{ N}$

2. angle  $= \tan^{-1} (3.9 \times 10^{-15} / 4.0 \times 10^{-15})$   
 $= 44^\circ$

[3]

(c) downward sloping line from (0, 2.0)

magnitude of gradient of line increases with time and line ends at (T, 0)

[2]

#### 42. O/N 17/P21/Q7

(a) (the ohm is) volt / ampere

[1]

(b) (i)  $R = \rho L / A$

$$\text{ratio} = [\rho L / (\pi d^2 / 4)] / [0.028 \rho \times 7.0 L / \{\pi (14d)^2 / 4\}] = 1000 \text{ or}$$

$$\text{ratio} = 14^2 / (0.028 \times 7) = 1000$$

[2]

(ii) same current (in connecting and filament wires) and  
the lamp/filament (wire) has greater resistance

[1]

(iii)  $P = V^2 / R$  or  $P = VI$  or  $P = I^2 R$

$$(\text{for filament wire}) R = 12^2 / 6.0 \text{ or } R = 6.0 / 0.50^2 \text{ or } R = 12 / 0.50$$

$$(\text{for filament wire}) R = 24 \Omega$$

$$(\text{for connecting wire}) R = 24 / 1000 = 2.4 \times 10^{-2} \Omega$$

[3]

(iv) resistance of connecting wire increases

current in circuit/lamp/filament (wire) decreases or

potential difference across lamp/filament (wire) decreases

(so) resistance of lamp/filament (wire) decreases

[3]

#### 43. O/N 17/P23/Q6

(a) (i)  $P = VI$

$$I = 30 / 120$$

$$= 0.25 \text{ A}$$

[2]

(ii)  $Q = 0.25 \times 3.0 \times 3600 (= 2700)$

$$\text{number} = (0.25 \times 3.0 \times 3600) / 1.60 \times 10^{-19}$$

$$= 1.7 \times 10^{22}$$

[2]

- (b)  $R = V/I$  or  $R = P/I^2$  or  $R = V^2/P$   
 $= 120/0.25$  or  $= 30/0.25^2$  or  $= 120^2/30 = 480 \Omega$  [2]
- (c)  $R = \rho l/A$   
 $A = (6.1 \times 10^{-7} \times 580 \times 10^{-3})/480 (= 7.37 \times 10^{-10})$   
 $d = [(4 \times 7.37 \times 10^{-10})/\pi]^{1/2}$   
 $= 3.1 \times 10^{-5} \text{ m}$  [3]
- (d) temperature decreases and so resistance decreases [1]

## 44. M/J 17/P21/Q6

- (a) volt/ampere [1]
- (b) (i)  $R_T = [1/3.0 + 1/6.0]^{-1} + 4.0 (= 6.0 \Omega)$   
 $I = 1.5/6.0$   
 $= 0.25 \text{ A}$  [3]
- (ii)  $V_B = 0.5 \text{ V}$   
 $I = 0.5/3.0$   
 $= 0.17 (0.167) \text{ A}$  [1]
- (iii)  $P = I^2 R$  or  $VI$  or  $V^2/R$   
 $\text{ratio} = (0.167^2 \times 3.0)/(0.25^2 \times 4.0)$   
 $= 0.33$  [2]
- (c) (i) vary/change/different radius/diameter/cross-sectional area (of wire) [1]
- (ii)  $v = I/Ane$   
 $\text{ratio} = \frac{(I_B/A_B)}{(I_C/A_C)} \text{ or } \frac{I_B}{I_C} \times \frac{A_C}{A_B}$   
 $(R \propto 1/A \text{ so}) \text{ ratio} = \frac{I_B}{I_C} \times \frac{R_B}{R_C} = \frac{0.167}{0.25} \times \frac{3.0}{4.0}$   
 $= 0.50$  [2]
- (d) (i) 0.25 A to 0.13 (0.125) A or halved [1]
- (ii) no change [1]

## 45. M/J 17/P23/Q6

- (a) (i) straight line through the origin [1]
- (ii) zero current for one direction (–ve V) up to zero or a few tenths of volt (+ve V)  
 straight line positive gradient/increasing gradient (+ve V) [2]
- (b) (i) 1. current = 2.8 A  
 2. 4(0) A for each lamp  
 current in circuit = 8(0) A [3]
- (ii) use of  $R = V/I$  with correct values of V from graph for each arrangement  
 1. series resistance ( $= 2.1 + 2.1$ ) = 4.2 or 4.3  $\Omega$   
 or  $(12/2.8) = 4.3 \Omega$   
 2. parallel resistance 1.5  $\Omega$  (each lamp 3.0  $\Omega$ )  
 or  $(12/8.0) = 1.5 \Omega$  [3]
- (iii) power =  $IV$  or  $V^2/R$  or  $I^2 R$   
 $\text{ratio} = (2.8 \times 6.0)/(4.0 \times 12) = 0.35$  [2]

## 46. O/N 16/P21/Q6

- (a) work done or energy (transformed) (from electrical to other forms)  
 charge [1]

- (b) (i) 1.  $V = IR$  or  $E = IR$   
 $I = 14/6.0$   
 $= 2.3 \text{ (2.33) A}$  [2]
2. total resistance of parallel resistors  $= 8.0 \Omega$   
current  $= 14/(6.0 + 8.0)$   
 $= 1.0$  [2]
- (ii)  $P = EI$  (allow  $P = VI$ ) or  $P = V^2/R$  or  $P = I^2R$   
change in power  $= (14 \times 2.33) - (14 \times 1.0)$   
or  $(14^2/6.0) - (14^2/14)$   
or  $(2.33^2 \times 6.0) - (1.0^2 \times 14)$   
 $= 19 \text{ W (18 W if 2.3 A used)}$  [2]
- (c)  $I = Anvq$   
ratio  $= (0.50n/n) \times (1.8 \text{ A/A})$  or ratio  $= 0.50 \times 1.8$   
 $= 0.90$  [2]

## 47. M/J 16/P21/Q6

- (a) (coulomb is) ampere second [1]
- (b) (total) charge or  $Q = nAle$   
 $I = Q/t$  and  $l/t = v$   
 $I = nAle/t = nAve$  therefore  $v = I/nAe$  [3]
- (c) (i) ratio  $= (I/nA_Ye)/(I/nA_Ze)$   
 $= A_Z/A_Y$  or  $4A/A$  or  $\pi d^2/(\pi d^2/4)$   
 $= 4$  [3]
- (ii)  $R = \rho l/A$  or  $R = 4\rho l/\pi d^2$   
 $R_Y = \rho l/A$  and  $R_Z = \rho(2l)/4A$  so  $R_Y/R_Z = 2$  or  
 $R_Y = 4\rho l/\pi d^2$  and  $R_Z = 4\rho(2l)/\pi 4d^2$  or  $2\rho l/\pi d^2$  so  $R_Y/R_Z = 2$  [2]
- (iii)  $V = 12R_Y/(R_Y + R_Z)$  or  $I = 12/(R_Y + R_Z)$  and  $V = IR_Y$   
 $V = 12 \times 2/3$   
 $= 8.0 \text{ V}$  [2]
- (iv) ratio  $= I^2R_Y/I^2R_Z$  or  $(V_Y^2/R_Y)/(V_Z^2/R_Z)$  or  $(V_YI)/(V_ZI)$   
 $= 2$  [1]

## 48. M/J 16/P23/Q6

- (a) ohm is volt per ampere or volt/ampere [1]
- (b) (i)  $R = \rho l/A$   
 $R_P = 4\rho(2l)/\pi d^2$  or  $8\rho l/\pi d^2$  or  $R_Q = \rho l/\pi d^2$  or  
ratio idea e.g. length is halved hence  $R$  halved and diameter is halved hence  $R$  is  $1/4$   
 $R_Q (= 4\rho l/\pi 4d^2) = \rho l/\pi d^2$   
 $= R_P/8$   
 $(= 12/8) = 1.5 \Omega$  [3]
- (ii) power  $= I^2R$  or  $V^2/R$  or  $VI$   
 $= (1.25)^2 \times 12 + (10)^2 \times 1.5$  or  $(15)^2/12 + (15)^2/1.5$  or  $15 \times 11.25$   
 $= (18.75 + 150) = 170 \text{ (168.75) W}$  [3]
- (iii)  $I_P = (15/12) = 1.25 \text{ (A)}$  and  $I_Q = (15/1.5) = 10 \text{ (A)}$   
 $V_P/V_Q = I_P n A_Q e / I_Q n A_P e$  or  $(1.25 \times \pi d^2)/(10 \times \pi d^2/4)$   
 $= 0.5$  [3]

**49. O/N 15/P22/Q6**

- (a) internal resistance causes lost volts  
p.d. across lamp is less than 12 V, power is less than 48 W [2]
- (b) (i) greater lost volts or p.d. across cell/lamp reduced, less current in lamp [1]  
(ii) p.d. across lamp/current in lamp decreases, hence resistance decreases [1]

**50. O/N 15/P23/Q5**

- (a) (i) resistance =  $V/I$   
very high/infinite resistance at low voltages  
resistance decreases as  $V$  increases [3]
- (ii) p.d. from graph 0.50 (V)  
resistance =  $0.5/(4.4 \times 10^{-3})$   
= 110 (114)  $\Omega$  [2]
- (b) (i) current (=  $1.2/375$ ) =  $3.2 \times 10^{-3}$  A [1]  
(ii) current in diode =  $4.4 \times 10^{-3}$  (A)  
total resistance =  $1.2/4.4 \times 10^{-3} = 272.7 (\Omega)$   
resistance of  $R_1 = 272.7 - 113.6 = 160$  (159)  $\Omega$  or  
p.d. across diode = 0.5 V and p.d. across  $R_1 = 0.7$  V  
resistance of  $R_1 = 0.7/4.4 \times 10^{-3}$   
= 160 (159)  $\Omega$  [2]
- (iii) power =  $IV$  or  $I^2R$  or  $V^2/R$   
ratio =  $(4.4 \times 0.5)/(3.2 \times 1.2)$   
or  $[(4.4)^2 \times 114]/[(3.2)^2 \times 375]$   
or  $[(0.5)^2 \times 375]/[114 \times (1.2)^2]$   
= 0.57 [2]

**51. M/J 15/P22/Q5**

- (a) curved line showing decreasing gradient with temperature rise  
smooth line not touching temperature axis not horizontal or vertical anywhere [2]
- (b) (i) (no energy lost in battery because) no/negligible internal resistance [2]  
(ii)  $I = V/R$   
=  $8/15 \times 10^3$  or  $1.6/3.0 \times 10^3$  or  $2.4/4.5 \times 10^3$  or  $12/22.5 \times 10^3$   
=  $0.53 \times 10^{-3}$  A [2]
- (iii) p.d. across X =  $12 - 8.0 - 3.0 \times 10^3 \times 0.53 \times 10^{-3}$  (= 2.4 V)  
 $R_X = 2.4/(0.53 \times 10^{-3})$  or  
 $R_{\text{tot}} = 12/0.53 \times 10^{-3}$  (=  $22.5 \times 10^3 \Omega$ )  
 $R_X = (22.5 - 15.0 - 3.0) \times 10^3$   
 $4.5(2) \times 10^3 \Omega$  [3]
- (iv) resistance decreases hence current (in circuit) is greater  
p.d. across X and Y is greater hence p.d. across Z decreases  
or explanation in terms of potential divider:  
 $R_Z$  decreases so  $R_Z/(R_X + R_Y + R_Z)$  is less  
therefore p.d. across Z decreases [2]

**52. M/J 15/P21/Q5**

- (a) very high/infinite resistance for negative voltages up to about 0.4 V  
resistance decreases from 0.4 V [2]

- (b) initial straight line from (0,0) into curve with decreasing gradient but not to horizontal repeated in negative quadrant [2]
- (c) (i)  $R = 12^2/36 = 4.0\ \Omega$  or  
 $I = P/V = 36/12 = 3.0\ \text{A}$  and  $R = 12/3.0 = 4.0\ \Omega$  [1]
- (ii) lost volts  $= 0.5 \times 2.8 = 1.4\ (\text{V})$  or  $E = 12 = 2.8 \times (R + r)$   
 $R = V/I = (12 - 1.4)/2.8$  or  $(R + r) = 4.29\ \Omega$   
 $= 3.8\ (3.79)\ \Omega$  or  $R = 3.8\ \Omega$  [3]
- (d) resistance of the lamp increases with increase of  $V$  or  $I$  [1]

**53. M/J 15/P23/Q5**

- (a)  $R = \rho l / A$   
 $= (5.1 \times 10^{-7} \times 0.50) / \pi(0.18 \times 10^{-3})^2 = 2.5\ (2.51)\ \Omega$  [2]
- (b) (i) resistance of CD  $= 8 \times$  resistance of AB  $= 20\ (\Omega)$   
circuit resistance  $= [1/5.0 + 1/20]^{-1} = 4.0\ (\Omega)$   
current  $= V/R = 6.0/4.0$   
 $= 1.5\ \text{A}$  [4]
- (ii) power in AB  $= I^2 R$  or power  $= V^2/R$   
 $= (1.2)^2 \times 2.5 = 3.6\ \text{W}$   $= (3.0)^2/2.5 = 3.6\ \text{W}$  [2]
- (iii) potential drop A to M  $= 1.25 \times 1.2 = 1.5\ \text{V}$   
potential drop C to N  $= 3.0\ \text{V}$   
p.d. MN  $= 1.5\ \text{V}$  [2]

**54. O/N 14/P21/Q3**

- (a) random error (in the measurements) of the length OR resistance [1]
- (b) gradient  $= (3.6 - 1.9) / (0.8 - 0.4)$   
 $= 4.25$  [2]
- (c)  $R = \rho l / A$   
 $\rho = \text{gradient} \times \text{area} = 4.25 \times 0.12 \times 10^{-6}$   
 $= 5.1(0) \times 10^{-7}\ \Omega\text{m}$  [3]
- (d) resistance decreasing with increasing area  
correct shape with curve being asymptote to both axes [2]

**55. M/J 14/P21/Q6**

- (a) e.m.f.: energy converted from chemical/other forms to electrical per unit charge  
p.d.: energy converted from electrical to other forms per unit charge [2]
- (b) (i) the p.d. across the lamp is less than 12 V  
or there are lost volts/power/energy in the battery/internal resistance [1]
- (ii)  $R = V^2/P$  (or  $V = RI$  and  $P = VI$ )  
 $= 144/48$   
 $= 3.0\ \Omega$  [2]
- (iii)  $I = E/(R_T + r)$   
 $= 12/2.0$   
 $= 6.0\ \text{A}$  [2]
- (iv) power of each lamp  $= I^2 R$   
 $= (3.0)^2 \times 3.0$   
 $= 27\ \text{W}$  [2]
- (c) less resistance (in circuit)/more current  
more lost volts/less p.d. across battery [2]